Idea Rents and Firm Growth

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Abstract

Which firms drive aggregate productivity growth? We document that firms with high price-earnings ratios tend to see increases in their subsequent earnings relative to sales, which we interpret as rents from ideas (innovation). We construct an endogenous growth model with shocks to firm innovation step-sizes and R&D efficiency and calibrate it to match patterns in the data. The model implies that growth would be much lower, even with the same innovative effort, if firms had the same step sizes. The model can be used to infer expected growth contributions of individual firms (such as members of the Magnificent Seven). We find that the share of growth coming from smaller listed firms substantially exceeds their sales share, whereas the largest listed firms account for less than their sales share.

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1 Introduction

How much do different firms contribute to aggregate productivity growth? Gibrat's Law says that, across firms, the growth rate of sales or employment does not correlate with size (Gibrat, 1931). Based on this empirical regularity, growth models commonly feature growth across firms that is iid or at least unrelated to firm size such as in the canonical Klette and Kortum (2004) model. A common implication of this property is that a firm's expected contribution to aggregate growth is proportional to its current sales share.

In contrast, Luttmer (2011) finds that iid firm growth cannot explain the young age of the largest firms in the U.S. He calculates that it should take firms centuries to become so large, whereas in the data the largest firms are much younger on average (e.g., Wal-Mart, Apple, or Amazon). Luttmer concludes some firms must have persistently high innovative opportunities.

In this paper we do three key things. First, we document that firms with high stock price-earnings (P/E) ratios exhibit faster subsequent earnings growth. This provides support for Luttmer's inference that firms must have predictably different growth rates. Second, we construct an endogenous growth model and calibrate it to match the empirical patterns. Third, we glean insights from our calibrated model about the expected contributions of different size firms to aggregate productivity growth.

On the empirical side, we use publicly-listed Compustat firms in all sectors from 1976 to 2024. In addition to faster earnings growth, higher P/E ratios predict faster growth in earnings *relative* to sales. This is true even when we look at earnings gross of R&D spending, which we can do for firms in manufacturing. We find that research intensity (R&D spending relative to sales) is increasing in P/E ratios and decreasing in firm sales, respectively. The former is consistent with high P/E firms being more innovative. The latter suggests that smaller Compustat firms are relatively more innovative. Intriguingly, we also find that sales growth is negatively related to sales. This contradiction of Gibrat's Law may partly reflect selection into the Compustat sample: for a firm with modest sales to go public, it may need to exhibit high growth potential.[1]

[1]The size and growth relationship may not all be due to selection. For all firms in the U.S. or for all U.S. manufacturers, including privately-held ones, Haltiwanger, Jarmin and Miranda (2011) and Akcigit and Kerr (2018) also find growth declining with size, where size is measured by employment.

Motivated by the facts we document, we construct an endogenous growth model in the spirit of Klette and Kortum (2004). Like their model, ours features multiproduct firms engaging in creative destruction through quality innovations. We deviate in three ways that are critical for mimicking the data: (i) stochastic idiosyncratic innovation step sizes across firms; (ii) stochastic R&D efficiency across firms; and (iii) innovation arrival rates that are a convex function of a firm's R&D expenditures alone.[2]

Step size shocks induce heterogeneity in P/E ratios, as price-cost markups are increasing in step sizes (Aghion and Howitt, 1992). Permanent differences in step sizes, however, would not generate heterogeneous P/E ratios in our model along a balanced growth path. Stochastic firm step sizes, in contrast, generate persistent but not permanent differences in P/E ratios across firms. High P/E ratio firms enjoy *temporarily* high step sizes, leading their sales to grow quickly for a while. In our model, high P/E ratios portend faster growth in earnings *relative* to sales. As firms add high-markup products, they increase the average markup in their portfolio of products. On top of this, the firm's future R&D spending will revert back down along with its innovation step size, boosting earnings net of R&D. Meanwhile, we find that differences in firm R&D efficiency are essential for generating realistic dispersion in firm size. The convexity of R&D costs otherwise constrains firms from differing dramatically in the level of their R&D investment and hence size in terms of sales (which are proportional to the number of products in the portfolio).

Klette and Kortum (2004) assumed that R&D costs are increasing in a firm's innovation arrival rate, but decreasing in the firm's stock of products. We drop the stock of products from the firm R&D cost function, akin to Akcigit and Kerr (2018), which makes it harder for larger firms to innovate. Doing so renders the model more tractable as a firm's number of products is no longer a state variable affecting its R&D decision. More importantly, it helps us explain why R&D/Sales ratios are falling in firm size among U.S. listed firms, and why high sales firms exhibit slower sales growth.

We calibrate our model to match 10 moments from the Compustat data: earnings growth per effective worker (a proxy for TFP growth); the average firm-specific

[2]Berlingieri, De Ridder, Lashkari and Rigo (2025) find that bursts of creative destruction are needed to fit the Pareto tail of firm size. They document such bursts of new product introductions in French manufacturing. We show that our idiosyncratic shocks to research costs and opportunities likewise generate a Pareto tail in the firm size distribution, as in the Compustat data.

discount rate from David, Schmid and Zeke (2022); the share of predictable earnings growth from P/E ratios that is due to earnings/sales growth across firms; the share of predictable earnings/sales growth from P/E ratios that is due to gross-of-R&D earnings/sales growth across firms; total earnings divided by total sales; the gap between the 90th and 10th percentiles of predicted earnings growth based on P/E differences across firms; the gap between the 90th and 10th percentiles of predicted gross-of-R&D earnings growth based on P/E differences across firms; the projection of R&D/Sales on log sales across manufacturing firms; the 90/50 ratio of sales among Compustat firms within years; and the dispersion of the ratio of price to earnings across Compustat firms within years.

We choose values for nine parameters to match the ten moments: the curvature of the R&D cost function (R&D spending as a function of the innovation arrival rate); the household's discount factor; the drift, persistence, and volatility of firm-specific innovation step sizes and R&D efficiency; and the correlation between shocks to step sizes and shocks to R&D efficiency.

Our estimates imply that firms differ persistently in their step sizes and R&D efficiency. In our calibrated model the largest listed firms, who collectively account for 10% of sales contribute less than 2% of aggregate TFP growth over the subsequent five years. Conversely, the smallest listed firms accounting for 10% of sales are responsible for over 60% of growth. We find that growth would be 93% lower (0.08% rather than 1.1% per year) if aggregate innovative effort were the same but all firms with the same R&D efficiency shared the same step size. Thus heterogeneous step sizes are essential for understanding aggregate growth. Eliminating step size heterogeneity for given R&D efficiency, however, reduces misallocation from markup dispersion and increases the level of TFP by 5%.

Related literature

Our approach complements research using patents and R&D. This literature has uncovered rich insights, but patenting and R&D are less common outside manufacturing and software. Even within manufacturing, the propensity to patent may be low and vary across firms—see Argente, Baslandze, Hanley and Moreira (2020) for a comparison of products to patents in U.S. consumer packaged goods, and Aghion, Bergeaud, Boppart, Klenow and Li (2026) for the propensity to patent innovations in France. The methodology in the current paper can be applied to all firms with earnings and valuations, regardless of sector, patenting, or even R&D reporting.

Like us, Kogan, Papanikolaou, Seru and Stoffman (2017) use stock prices to gauge the value of firm innovations. They examine changes in firm stock market capitalization in response to the announcement of firm patent grants. Our effort is distinct in not relying on patents and in focusing on the expected future growth contributions of firms rather than valuing individual innovations. Our paper relates to studies of Tobin's Q, such as Hall (1993), Hall (2001), and Crouzet and Eberly (2023). Tobin's Q arguably reflects rents from both past and future innovation, whereas the P/E ratio highlights the discounted value of future innovations (and hence future growth contributions) not reflected in current earnings.

A vast literature explores the empirics of firm dynamics, in particular employment. Recent examples include Decker, Haltiwanger, Jarmin and Miranda (2016), Garcia-Macia, Hsieh and Klenow (2019), Sterk, Sedláček and Pugsley (2021), and Klenow and Li (2021). Our approach looks at earnings data to get at innovation rates and hence growth contributions. We find that fast-growing firms, also known as gazelles or Luttmer Rockets, are associated with rising earnings relative to sales in addition to growing employment.

Studies using U.S. data that include private firms (not just publicly-listed firms) uncover systematic deviations from Gibrat's Law. Dunne, Roberts and Samuelson (1989), Haltiwanger, Jarmin and Miranda (2011), and Akcigit and Kerr (2018) find that firm or plant growth declines with size in the Census of Manufactures and the Longitudinal Business Database (LBD) from the U.S. Census Bureau. Neumark, Wall and Zhang (2011) report the same qualitative pattern in the National Establishment Time Series data. Yeh (2025) finds that the variance of firm growth declines with size in the LBD. We contribute to this area by documenting deviations from Gibrat's Law for publicly-listed U.S. firms and quantifying the economic significance of the deviation in terms of expected growth contributions.

Recent precursors to our quantitative modeling of firm dynamics and growth are Cao, Hyatt, Mukoyama and Sager (2017), Akcigit and Kerr (2018), Acemoglu, Akcigit, Alp, Bloom and Kerr (2018), Akcigit and Ates (2023), and Peters and Walsh (2025). Cao et al. model stochastic R&D efficiency (but not stochastic step sizes) in a model with variety creation and own-product innovation by incumbents. The middle three studies focus on firms who conduct R&D and patent. Like us, Akcigit and Kerr (2018) entertain stochastic step sizes. Peters and Walsh aim to understand the impact of declining population growth on firm dynamism and productivity growth.

Campbell and Shiller (1988) and Cochrane (2011) decompose variation in stock market-wide P/E ratios into time variation in the stochastic discount factor versus growth in earnings (or cash flow). Similarly, Atkeson, Heathcote and Perri (2022) and Atkeson, Heathcote and Perri (2024) stress that fluctuating rents from innovation show up in aggregate P/E ratio movements. They explore the implications of this for time-varying U.S. stock market valuations and the U.S. net international asset position. In contrast to these studies, we investigate the implications of cross-firm dispersion in P/E ratios for firm growth contributions.

The rest of the paper is organized as follows. Section 2 documents empirical patterns in Compustat. Section 3 builds an endogenous growth model to help interpret these patterns. Section 4 calibrates the model and characterizes firm growth contributions. Section 5 concludes.

2 Empirical patterns

2.1 Data source

Our main variables are stock prices from the Center for Research in Security Prices (CRSP), "street" earnings from the Institutional Brokers' Estimate System (I/B/E/S), and sales from Compustat.[3] Our data cover 15,768 unique firms and 165,007 firm-year observations from 1976 to 2024.[4] While the data covers only publicly-listed firms, these firms account for roughly one-fourth of total private non-farm employment.[5]

[3]We use street earnings instead of Generally Accepted Accounting Principles (GAAP) earnings because the former is conceptually closer to firm performance and, according to Hillenbrand and McCarthy (2025), tracks aggregate GAAP earnings before deducting special items such as natural disasters. See Bradshaw and Sloan (2002) for more discussion.

[4]These are the first and last years when data is available.

[5]This is based on employment reported for Compustat firms relative to total private nonfarm employment from the BLS.

We source the security-level share price on the last trading date of the firm's fiscal year from CRSP. We multiply each firm's stock price by its number of outstanding shares to arrive at each firm's market capitalization. Our data covers securities listed on the NYSE, NYSE MKT, and NASDAQ exchanges. We extract security-level fiscal year-end earnings per share from I/B/E/S. The earnings per share (EPS) is "street-level" in that I/B/E/S adjusts the accounting EPS for items that the majority of analysts include or exclude from their valuation calculations. We construct firm-level earnings by multiplying EPS by the number of shares outstanding. We calculate the P/E ratio as firm-level market capitalization divided by firm-level earnings. We also use fiscal year sales and R&D data from Compustat.

We construct the growth of earnings h years ahead g_E^h and growth of the earnings-to-sales ratio $g_{E/S}^h$ as

$$g_E^h = \frac{1}{h}\log\left(\frac{E_{t+h}}{E_t}\right), \quad \text{and} \quad g_{E/S}^h = \frac{1}{h}\log\left(\frac{E_{t+h}/S_{t+h}}{E_t/S_t}\right).$$

We drop firms with negative earnings when calculating the growth rate of earnings or the earnings-to-sales ratio. Later on, we do the same in the model for purposes of comparing data and model moments.

2.2 Motivating facts

Here we document several patterns in the data that will motivate our modeling. First, firms with high P/E ratios tend to have high subsequent earnings growth—even relative to their sales growth. Second, within manufacturing where R&D data is available and more reliable, high P/E ratio firms exhibit higher R&D/Sales ratios. Third, firms with higher sales tend to have lower R&D/Sales and lower subsequent sales growth.

High P/E firms have higher subsequent growth of earnings and earnings/sales It is well known that the P/E ratio varies significantly across firms (see the Appendix Section A.1 discussion). This dispersion does not appear to be simply noise. Figure 1 plots the 5-year ahead earnings growth against the current-year price-to-earnings ratio across firms as a binned scatterplot. Firm-specific discount rates contribute to P/E variation across firms (De La O, Han and Myers, 2025) and may correlate with earnings growth.[6] We residualize both variables against firm-specific discount rates to show that P/E ratios predict earnings growth conditional on firm discount rates.

Specifically, both $\log$ P/E and earnings growth are regressed on a cubic polynomial of the firm-specific discount rate r_{it}, which we construct by adding the risk-free rate to the firm's risk premium estimated following David, Schmid and Zeke (2022) using the Fama-French three-factor model.[7] We also control for year and fiscal-year end month fixed effects because we are interested in the relationship across firms rather than aggregates over time. Thus, Figure 1 plots the residualized variables from the regression

$$\log P/E_{it} = \beta_0 + \beta_1 r_{it} + \beta_2 r_{it}^2 + \beta_3 r_{it}^3 + \tau_{y(t)} + \mu_{m(t)} + \varepsilon_{i,t}, \tag{1}$$

where $\tau_{y(t)}$ is a year fixed effect, and $\mu_{m(t)}$ a fiscal-year end month fixed effect; the earnings-growth residuals come from the analogous regression with earnings growth on the left-hand side. A high P/E ratio today predicts high subsequent earnings growth. The R^2 of this relationship is 19%.

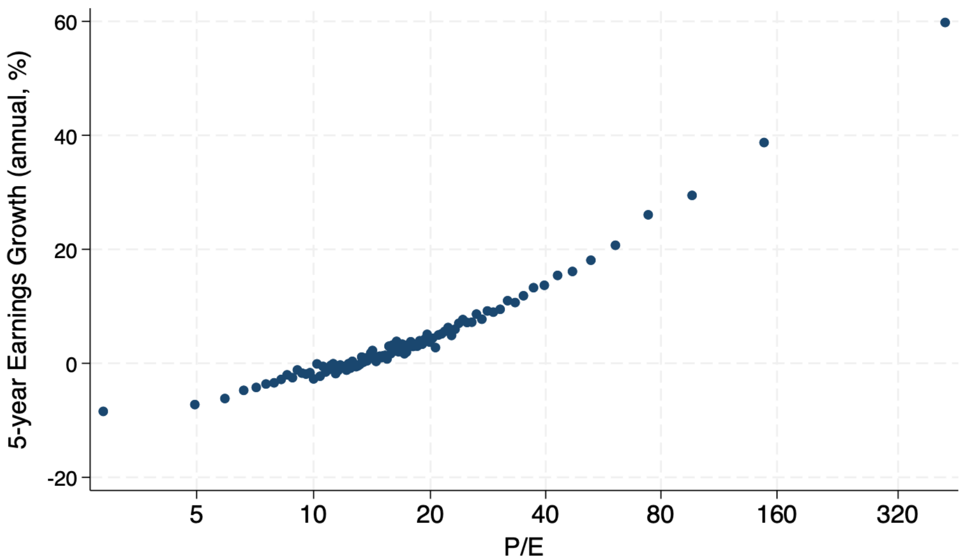
Earnings in the data likely deduct intangible investment such as R&D. As a result, P/E may predict earnings growth because firms that do more intangible investment today have lower measured E and higher E growth going forward (when the investment pays off). We ran regression (1) replacing $\log$ P/E with $\log$ P/(E+R&D) to remove movements in earnings due to subtracting R&D off earnings E. We do this for manufacturing firms where R&D is better measured. Figure 2 plots the residualized earnings growth g_E against P/(E+R&D) in the regression. It shows that the P/(E+R&D) ratio predicts future earnings growth.

Column 1 of Table 1 shows the coefficient from regressing 5-year ahead annual earnings growth on the current P/(E+R&D) ratio. The fitted values from this regression indicate significant dispersion in predicted earnings growth: the 90–10 percentile range of predicted earnings growth is 9 percentage points and 15 percentage points for growth of earnings before deducting R&D.

[6]For example, more innovative firms may enjoy a lower discount rate because they are rich in growth options and therefore better-hedged against systematic obsolescence shocks (Kogan and Papanikolaou, 2014).

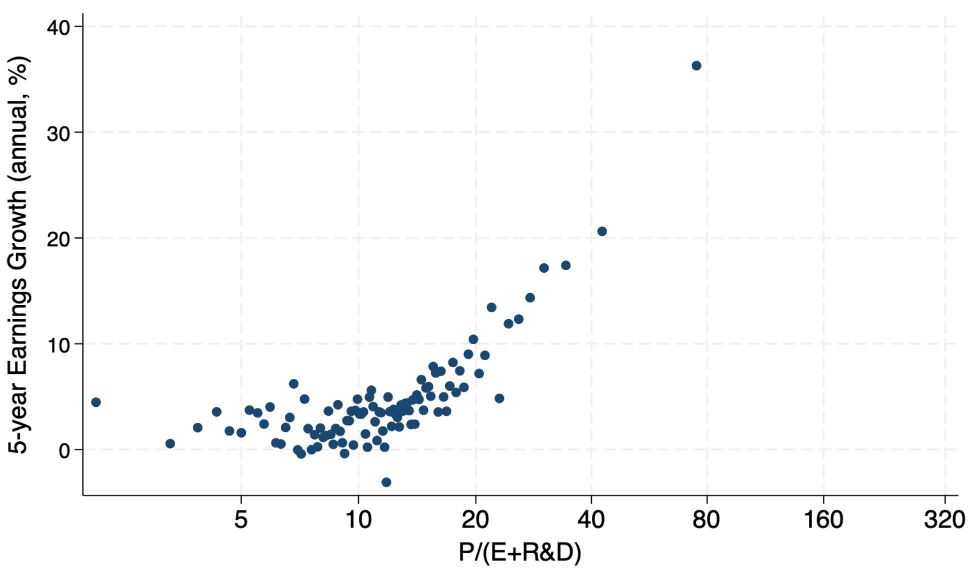
[7]Figure A5 plots the distribution of P/E across firms after residualizing on r_{it}.

Figure 1: Earnings growth vs. P/E, all industry



Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the P/E ratio in year t. The y-axis is average yearly earnings growth from year t to $t + 5$. Both variables are residualized on a cubic of the firm-specific discount rate from David et al. (2022), fiscal-year end month and year fixed effects. The R^2 of the underlying regression is 19%.

Figure 2: Earnings growth vs. P/(E+R&D), manufacturing



Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the $P/(E + R\&D)$ ratio in year t. The y-axis is average yearly earnings growth from year t to $t + 5$. Both variables are residualized on a cubic of the firm-specific discount rate from David et al. (2022), fiscal-year end month and year fixed effects. Manufacturing firms only.

Markups (or profitability per $ of sales) appear to play an important role in the relationship between earnings growth and P/E ratios in Figures 1 and 2. Columns 2, 3, and 4 of Table 1 report growth of earnings before R&D, growth of earnings before R&D relative to sales, and sales growth, respectively. A little under half of the relationship ((0.0690-0.0394)/0.0690 = 43%) between earnings growth and P/(E+R&D) is accounted for by higher P/E ratios predicting higher growth in earnings *relative* to sales $g_{E/S}$ as opposed to predicting just higher sales growth g_S. The contribution of $g_{E/S}$ (0.0283/(0.0690-0.0394) = 96%) comes mainly from higher P/(E+R&D) predicting higher growth of profitability $g_{(E+R\&D)/S}$ as opposed to movements in R&D $g_{E/(E+R\&D)}$ which motivates an operative markup channel.[8]

Table 1: Growth of earnings and earnings/sales in manufacturing

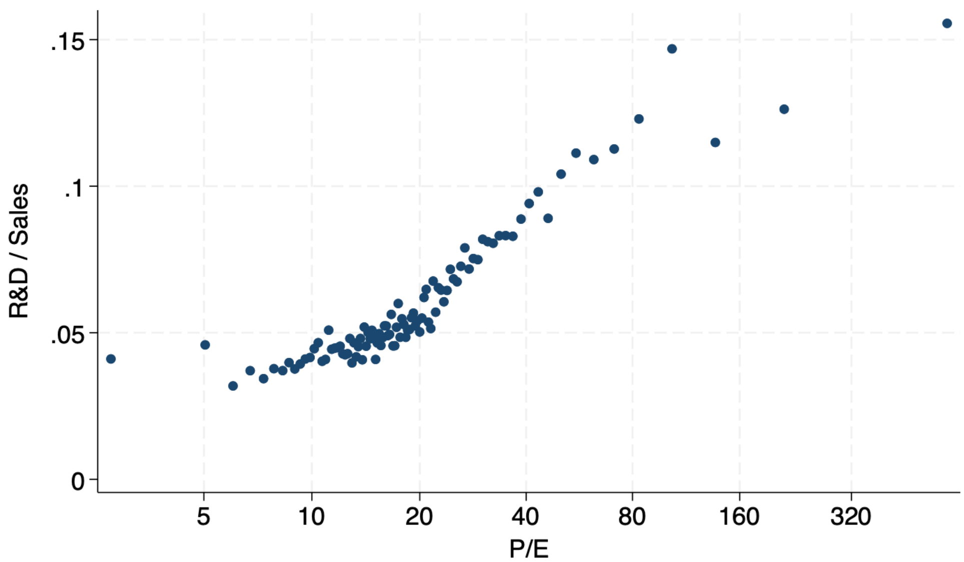
	(1)	(2)	(3)	(4)
	g_E	$g_{E+R\&D}$	$g_{(E+R\&D)/S}$	g_S
$\log(P/(E+R\&D))$	0.0690	0.0677	0.0283	0.0394
	(0.0073)	(0.0033)	(0.0023)	(0.0026)
N	18,644	18,644	18,644	18,644
R^2	0.040	0.107	0.063	0.093

Notes: Data from CRSP, I/B/E/S, and Compustat. Manufacturing firms only. Column (1) displays the coefficient when regressing average yearly earnings growth from year t to $t+5$ on $\log$ P/(E+R&D) ratio in t. Column (2) displays the coefficient when regressing average yearly earnings growth gross of R&D from year t to $t+5$ on $\log$ P/(E+R&D) ratio in t. Columns (3) and (4) decompose the coefficient from Column (2) into growth in the ratio of earnings gross of R&D to sales and growth in sales. All regressions control for year and fiscal-year end month fixed effects, as well as a cubic in firm-specific discount rate from David et al. (2022).

Firms with higher P/E ratios exhibit higher R&D intensity The predictability of earnings and profitability growth by P/E ratios may be related to innovations. Among manufacturing firms for whom R&D is reliably reported, high P/E firms have higher

[8] Appendix Table A1 shows the pattern holds at all horizons $h = 1, 2, \ldots, 5$. Appendix Table A2 shows the pattern is even stronger when using $\log$ P/E and including non-manufacturing firms: $g_{E/S}$ accounts for 78% of the relationship between g_E and $\log$ P/E.

Figure 3: R&D/Sales vs. P/E, manufacturing



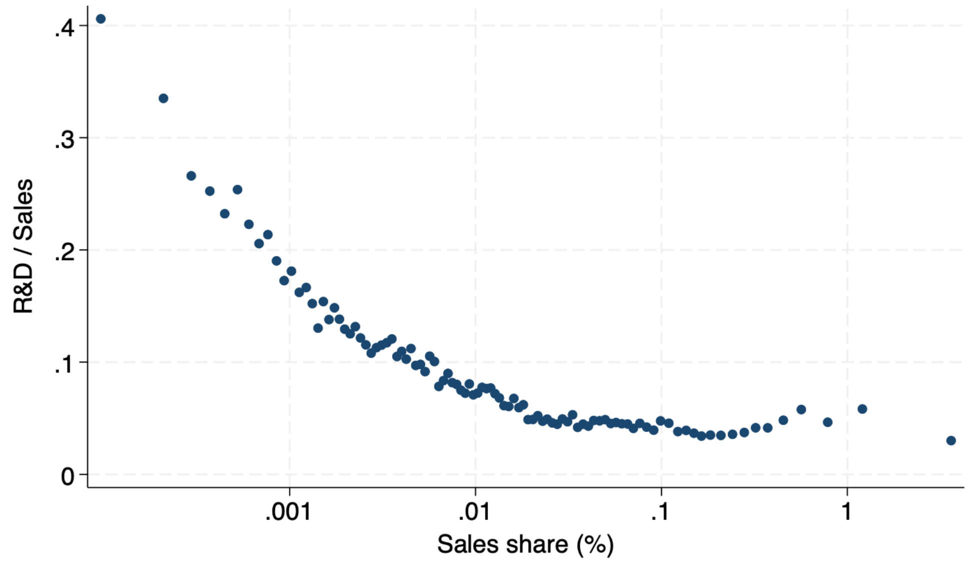
Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the P/E ratio in year t. The y-axis is R&D over sales in year t. Both variables are residualized on a cubic of the firm-specific discount rate from David et al. (2022), fiscal-year end month and year fixed effects. Manufacturing firms only.

R&D intensity, as presented in Figure 3.[9]

Firms with higher sales have lower R&D intensity and lower sales growth High sales firms tend to have lower R&D intensity, as displayed in Figure 4. Larger firms also have lower sales growth, as shown in Figure 5 for manufacturing firms and in Appendix Figure A6 for all firms. This empirical deviation from Gibrat's Law may partially reflect selection into the Compustat sample as small firms with high growth potential may be more likely to go public. That said, Akcigit and Kerr (2018) show that employment growth and patenting per employee decline with firm size within the sample of patenting firms in the LBD, which covers both public and private firms.

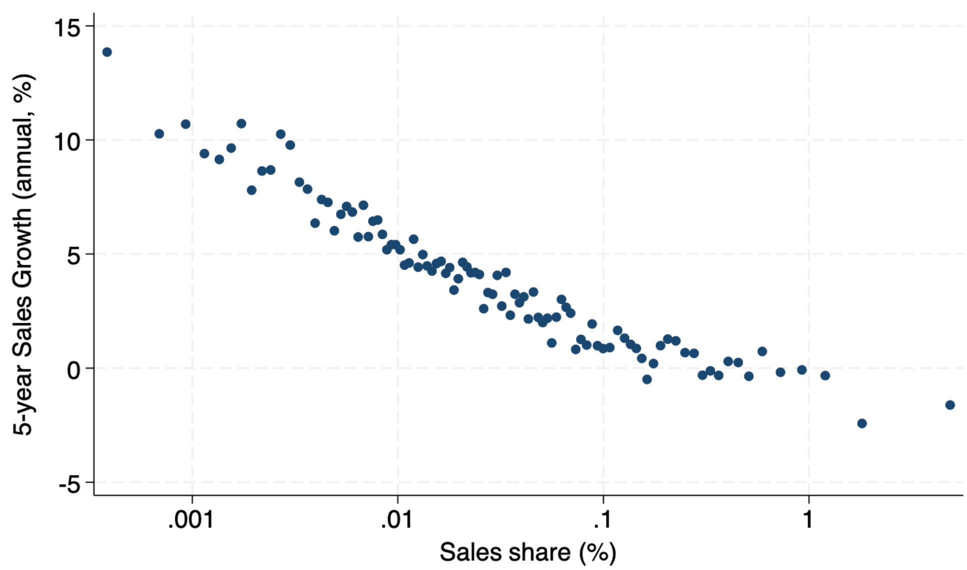
[9]The x-axis of Figure 3 is P/E and not $P/(E + R\&D)$ to avoid measurement errors in R&D driving a mechanical negative relationship between R&D intensity and the P/E ratio.

Figure 4: R&D/Sales vs. sales, manufacturing



Notes: Data from Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is sales relative to total sales across firms in t. The y-axis is R&D over sales in year t. Manufacturing firms only.

Figure 5: Sales growth vs. Sales, manufacturing



Notes: Data from Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is sales relative to total sales across firms in t. The y-axis is average yearly growth of sales from year t to year $t + 5$. Manufacturing firms only.

2.3 Robustness checks

In the next section, we pursue a theory of idiosyncratic innovation shocks to explain the motivating facts. We are motivated to consider such shocks because the following robustness checks show that plausible alternative explanations cannot fully explain the facts. In Appendix Table A6, we also show that the calibration target moments that we will use later are robust.

Firm age Young firms grow faster on average (Haltiwanger et al., 2011; Peters and Walsh, 2025) so that firm age differences could be driving the patterns above. In Appendix A.2, however, we show that our motivating facts hold in the sample of firms that have been in Compustat for at least ten years.

Physical capital and debt For simplicity and transparency, we use the unadjusted street earnings and market capitalization. In Appendix A.3 we show that the facts are robust to adjusting earnings and market capitalization for debt and tangible capital to be closer to rents and the market value of rents.

Firm fixed effects Appendix A.4 shows that our facts are robust to controlling for firm fixed effects instead of a measure of firm-specific discount rates. Firm fixed effects serve as a general proxy for firm-specific risk. They could also capture other permanent differences across firms such as differences in their customer base. We do not use this specification as the baseline because it potentially removes P/E ratio differences due to shocks rather than permanent differences for firms with fewer observations in the sample.

3 Model

Based on the facts we documented in the previous section, we formulate an endogenous growth model with stochastic but persistent differences in innovation step sizes and R&D efficiencies across firms. Growth in earnings/sales (especially earnings gross of R&D expenditures) is suggestive of rising markups at high P/E firms. Thus high P/E ratio firms may have opportunities to invest in innovations with high quality steps that allow them to command high markups, fueling their fast earnings growth. A high P/E ratio could then reflect that investors have information about the firm's investment opportunities that they price into firm valuations. If firms have information about their opportunities as well, this could account for the high R&D intensity of high P/E ratio firms. Stochastic R&D efficiency can also make sales growth increasing in P/E ratios because firms with temporarily good R&D efficiency spend more on R&D, which lowers earnings today but expands future sales.

Our model builds on Klette and Kortum (2004), which features endogenous quality growth and creative destruction by multiproduct firms. The key difference is that we allow each firm's innovation step size to evolve stochastically (and idiosyncratically) rather than being common or fixed over time. This will create markup dynamics and in turn heterogeneous P/E ratios to help the model match the patterns we found in the data. Because our motivating facts hold for incumbent firms, we abstract from firm entry and exit for simplicity.

3.1 Model setup

We consider a closed economy wherein time is discrete.

Household A representative household inelastically supplies L units of production labor and maximizes its discounted utility

$$\max_{\{C_t, A_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \log(C_t),$$

subject to

$$A_{t+1} = A_t(1+r_t) + w_t L - C_t,$$

and a standard no-Ponzi game condition. Here C denotes consumption, A wealth from owning the firms in the economy, r the interest rate, w the wage, and $\beta \in (0,1)$ the discount factor. The Euler equation resulting from the household problem links the equilibrium consumption growth rate to the discount factor and interest rate, as usual:

$$\frac{C_{t+1}}{C_t} = \beta(1+r_{t+1}). \tag{2}$$

Final goods production Final output Y is a Cobb-Douglas bundle of a unit interval of intermediate goods of qualities $q(i)$:

$$Y = \exp\left(\int_0^1 \log\left(q(i)y(i)\right) di\right), \tag{3}$$

where we suppressed time subscripts to ease notation. Final output can be used for consumption C or research expenditures R (so that $Y = C + R$).

We assume the final goods market to be competitive. Thus final goods production can be characterized by a representative firm solving

$$\max_{y(i),\, i\in[0,1]} \mathcal{P} \cdot \exp\left(\int_0^1 \log\left(q(i)y(i)\right) di\right) - \int_0^1 p(i)y(i)\, di, \tag{4}$$

taking prices $\mathcal{P}$ and $p(i)$ as given. Due to the Cobb-Douglas structure, the solution to the final goods producer's problem features equal spending on each product

$$p(i)y(i) = \mathcal{P}Y. \tag{5}$$

In the following, we normalize the price of the final good $\mathcal{P} = \exp\left(\int_0^1 \log\left(p(i)/q(i)\right) di\right)$ to 1 in all periods.

Intermediate goods production and quality innovations Intermediate goods are produced by a (large) fixed number of firms J indexed by $j \in \{1, \ldots, J\}$. Firm j has a blueprint to produce varieties i at quality $q(i,j)$. Firms need one unit of production labor to produce one unit of an intermediate good. More formally, firm j can produce one unit of intermediate good i at quality $q(i,j)$ with technology $y(i,j) = l(i,j)$ where $l(i,j)$ is the labor input of firm j into producing good i.

The quality levels at which a firm produces change over time as a result of R&D activities. Each firm has access to an R&D technology with heterogeneous and stochastic step sizes γ and R&D efficiency φ. That is, in period t a firm j can randomly draw x_j number of lines to innovate upon by spending

$$R_{j,t} = \frac{1}{\varphi_{j,t}} x_{j,t}^{\nu} Y_t \tag{6}$$

units of final output. The term $1/\varphi_{jt}$ captures an R&D cost shifter of firm j at time t and we refer to φ as R&D efficiency. We have $\nu > 1$. For the $x_{j,t}$ lines drawn by firm j in period t, the firms can improve the highest quality in the line by a factor of $\gamma_{j,t} > 1$ and can produce at this higher quality level from next period $t+1$ onward. In each period, the firms also lose products at rate $\bar{x}_t$, which is endogenous in general equilibrium but is taken as given by the firms.

The step size and R&D efficiency of a firm $(\gamma_{j,t}, \varphi_{j,t})$ follow an N-state Markov chain. We use index k for a distinct combination of (γ, φ) and we have N possible combinations with transition probabilities given by

$$M = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1k'} & \cdots & m_{1N} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ m_{N1} & m_{N2} & \cdots & m_{Nk'} & \cdots & m_{NN} \end{bmatrix}, \tag{7}$$

where $m_{kk'}$ denotes the probability of a firm with a step size and R&D efficiency combination k in t moving to a step size and R&D efficiency combination k' in $t+1$. The Markov chain is ergodic.

We assume that intermediate input markets feature Bertrand competition. Under this, the highest quality producer in a line i, $j(i)$ sets the quality-adjusted price to the quality-adjusted marginal cost of the second-highest quality producer $p(i) = \frac{w}{q(i,j'(i))}$ while its own quality-adjusted marginal cost is $\frac{w}{q(i,j(i))}$. Hence, the markup μ in a line is given by the quality of the best producer relative to the second-best producer:

$$\mu(i) = \frac{q(i, j(i))}{q(i, j'(i))}. \tag{8}$$

Since the highest-quality producer is the firm that last innovated on the line, the markup in a line is determined by the quality step size of the last innovator at the time it acquired the line. In general this step size differs from the step size at which the producer currently innovates. In the following, we will denote the step size of the last innovation (and therefore the equilibrium markup) on line i by $\gamma(i)$. With this markup the profit relative to aggregate output Y of the firm producing in line i is[10]

$$\pi(\gamma(i)) \;=\; 1 - \frac{1}{\gamma(i)}. \tag{9}$$

Let $\mathcal{N}_j$ be the portfolio of lines where firm j is the highest-quality producer. The measure of $\mathcal{N}_j$ is n_j and we call this the number of lines of firm j. Firm j's total earnings before expensing R&D, expressed relative to output Y, are $\int_{i\in\mathcal{N}_j} \pi(\gamma(i))\, di$. Because there are a finite number of possible step sizes which we can index by ι, we can write these earnings gross of R&D (again expressed relative to Y) as

$$\tilde{E}(\{\omega_{j\iota}\}_\iota, n_j) = n_j \sum_\iota \omega_{j\iota}\pi(\gamma_\iota), \tag{10}$$

where $\omega_{j\iota}$ is the share of firm j's portfolio where the firm had step size γ_ι when it acquired the line and n_j is the number of lines in the portfolio. We have $\sum_\iota \omega_{j\iota} = 1$. Earnings after R&D (expressed relative to aggregate output) are then

$$E(\{\omega_{j\iota}\}_\iota, n_j, x_j, \varphi_j) = \tilde{E}(\{\omega_{j\iota}\}_\iota, n_j) - \frac{1}{\varphi_j} x_j^\nu. \tag{11}$$

In each period, the firm sees its current step size γ_{jt} and R&D efficiency φ_{jt} and then chooses the number of lines to innovate upon x_j to maximize the value of the firm. The R&D cost is paid in period t while the newly acquired lines enter the firm's portfolio in $t+1$. The value of the firm (before dividends) is the expected present discounted value of its earnings stream

$$\begin{aligned} \mathcal{V}_{j,t} \;=\; & \max_{\{x_{j,t+\tau}, n_{j,t+\tau+1}\}_{\tau=0}^\infty} \Bigg\{ E\left(\{\omega_{j\iota,t}\}_\iota, n_{j,t}, x_{j,t}, \varphi_{jt}\right) Y_t \\ & + \mathbb{E}_t \sum_{\tau=1}^\infty \left(\prod_{s=1}^\tau \frac{1}{1+r_{t+s}} \right) E\left(\{\omega_{j\iota,t+\tau}\}_\iota, n_{j,t+\tau}, x_{j,t+\tau}, \varphi_{j,t+\tau}\right) Y_{t+\tau} \Bigg\}, \end{aligned} \tag{12}$$

subject to

$$E\left(\{\omega_{j\iota,t+\tau}\}_\iota, n_{j,t+\tau}, x_{j,t+\tau}, \varphi_{j,t+\tau}\right) = n_{j,t+\tau} \sum_\iota \omega_{j\iota,t+\tau} \left(1 - \frac{1}{\gamma_\iota}\right) - \frac{1}{\varphi_{j,t+\tau}} x_{j,t+\tau}^\nu, \tag{13}$$

[10]To prepare an analysis in detrended variables below we already express profits and earnings relative to total output Y.

for all τ and

$$\omega_{j\iota,t+\tau+1} n_{j,t+\tau+1} = (1-\bar{x}_{t+\tau})\omega_{j\iota,t+\tau} n_{j,t+\tau} + \mathbf{1}\{\gamma_{j,t+\tau} = \gamma_\iota\} x_{j,t+\tau}, \quad \forall \tau, \iota. \tag{14}$$

Equation (14) keeps track of the changes in the product portfolio of a firm across the different markup types (indexed by ι), whereas (13) restates the expression for earnings. Together with (11), (13) and (14) imply the following dynamics for earnings gross of R&D expenses:

$$\tilde{E}(\{\omega_{j\iota,t+\tau+1}\}_\iota, n_{j,t+\tau+1}) = (1-\bar{x}_{t+\tau})\tilde{E}(\{\omega_{j\iota,t+\tau}\}_\iota, n_{j,t+\tau}) + x_{j,t+\tau}\pi(\gamma_{j,t+\tau}), \ \forall \tau. \tag{15}$$

Earnings in a given period depend on the history of a firm's past innovations and on how fast the firm loses its products due to creative destruction. Firms which are large today in terms of earnings must have had a history of high innovation activities x. Even conditional on the innovation activity, firms with a higher innovation step see their earnings grow faster as they add more profitable lines.

Market clearing and aggregates Since we have a unit measure of lines, the aggregate rate of creative destruction is equal to

$$\bar{x}_t = \sum_{j=1}^{J} x_{j,t}. \tag{16}$$

We also have for all t the labor market clearing condition

$$\int_0^1 l_t(i)di = L,$$

where $l_t(i)$ is the labor input used to produce line i, the asset market clearing condition

$$\sum_{j=1}^{J} \mathcal{V}_{j,t} = A_t,$$

and goods market clearing

$$\sum_{j=1}^{J} R_{j,t} = R_t = Y_t - C_t, \tag{17}$$

where total output is given by (3). Finally, we have the number (fraction) of lines operated by all the firms summing to 1, i.e.,

$$1 = \sum_{j=1}^{J} n_{j,t}.$$

BGP equilibrium definition A decentralized equilibrium consists of a sequence of quantities and prices that jointly solve the problems of the final producer, the intermediate producer, the household and are consistent with the market clearing conditions and the aggregate constraints stated above.

In the following we focus on a balanced growth path (BGP) which we define the standard way, i.e., as a path along which all quantities grow at constant rates. Along the BGP there is an ergodic joint distribution of firm size, firm states (γ, φ), and markups. Furthermore, the interest rate as well as the rate of creative destruction are constant and the growth rate of consumption equals the growth rate of output and we denote them by $r^\star$, $\bar{x}^\star$ and $C_{t+1}/C_t - 1 = Y_{t+1}/Y_t - 1 = g^\star$, respectively, along a BGP. We can then use (16) to express

$$\bar{x}^\star = \sum_{\iota=1}^{N} J s_\iota x(\gamma_\iota, \varphi_\iota), \tag{18}$$

where s_ι is the stationary share of firms with a particular combination of step size γ and R&D efficiency φ consistent with the Markov transition probabilities in the matrix M, and $x(\gamma_\iota, \varphi_\iota)$ denotes the privately optimal R&D policy along the BGP given this (γ, φ) combination. As there is no change in allocative efficiency along a BGP, all output growth comes from quality improvements and we have

$$1 + g^\star = \prod_{\iota=1}^{N} (\gamma_\iota)^{J s_\iota x(\gamma_\iota, \varphi_\iota)}. \tag{19}$$

Note that, conditional on their innovation arrival rate x, high step size firms contribute more to growth. Finally, the Euler equation (2) links the growth rate of the economy to the interest rate in the usual way:

$$1 + g^\star = \beta(1 + r^\star). \tag{20}$$

Firm problem along the balanced growth path Along a BGP the dynamic firm problem simplifies to

$$V(\{\omega_{j\iota,t}\}_\iota, n_{j,t}, \gamma_{jt}, \varphi_{j,t}) = \tilde{E}(\{\omega_{j\iota,t}\}_\iota, n_{j,t}) \frac{1}{1-\beta(1-\bar{x}^\star)} + \max_{\{x_{j,t+\tau}\}_{\tau=0}^{\infty}} \mathbb{E}_t \left[\sum_{\tau=0}^{\infty} \beta^\tau \left(\beta x_{j,t+\tau} v(\gamma_{j,t+\tau}) - \frac{1}{\varphi_{j,t+\tau}} x_{j,t+\tau}^\nu \right) \right], \tag{21}$$

where $v(\cdot)$ denotes the present discounted value of a line acquired with step size γ (expressed relative to output Y). Formally,

$$v(\gamma_{j,t+\tau}) = \frac{1 - \frac{1}{\gamma_{j,t+\tau}}}{1-\beta(1-\bar{x}^\star)}, \tag{22}$$

and $V(\cdot)$ is the value of a firm relative to output, i.e., $V(\cdot) = \mathcal{V}(\cdot)/Y$. The right-hand side of the first line in (21) captures the value from the portfolio of products that the firm acquired by past innovations, whereas the second line is the firm value from expected earnings from new innovations in the future. This expectation is only conditional on the current step size and R&D efficiency because these states follow a Markov process. In sum, the firm states become the current γ and φ as well as the number of products n acquired by past innovations and their markup composition $\{\omega_\iota\}_\iota$.

In the firm problem above, the optimal choice of $x_{j,t}$ only depends on the current $\gamma_{j,t}$ and $\varphi_{j,t}$. Hence, we can simplify the R&D decision as choosing a policy function $x(\gamma, \varphi)$ that solves

$$\max_{x \geq 0} \left\{ \beta x v(\gamma) - \frac{1}{\varphi} x^\nu \right\}. \tag{23}$$

The solution to this maximization problem is

$$x(\gamma, \varphi) = \left(\beta v(\gamma) \frac{\varphi}{\nu} \right)^{1/(\nu-1)}, \tag{24}$$

which says that firms with higher current step sizes γ and higher R&D efficiency φ choose a higher innovation activity x. A high quality step and high R&D efficiency both present a window of opportunity with superior innovation technology. Equilibrium

R&D spending (relative to output) is then

$$\frac{R(\gamma,\varphi)}{Y} = \frac{1}{\varphi}x(\gamma,\varphi)^{\nu} = \varphi^{\frac{1}{\nu-1}}\left(v(\gamma)\frac{\beta}{\nu}\right)^{\nu/(\nu-1)}, \tag{25}$$

which increases in the step size and R&D efficiency. The additional firm rent generated by today's R&D (net of the R&D cost) is

$$(\beta v(\gamma))^{\nu/(\nu-1)} \cdot \left(1-\frac{1}{\nu}\right)\left(\frac{\varphi}{\nu}\right)^{1/(\nu-1)}. \tag{26}$$

Hence, firms with higher step sizes and higher R&D efficiency innovate more, spend more on R&D, and grow the firm value by more.

Substituting the privately optimal R&D activity (24) into the firm's value function yields the (detrended) equilibrium value of a firm along the BGP:

$$\begin{aligned} V\left(\{\omega_{j\iota,t}\}_{\iota}, n_{j,t}, \gamma_{j,t}, \varphi_{j,t}\right) &= \tilde{E}(\{\omega_{j\iota,t}\}_{\iota}, n_{j,t})\frac{1}{1-\beta(1-\bar{x})} \\ + & \left(1-\frac{1}{\nu}\right)(\beta v(\gamma_{j,t}))^{\nu/(\nu-1)}\left(\frac{\varphi_{j,t}}{\nu}\right)^{1/(\nu-1)} \\ + & \left(1-\frac{1}{\nu}\right)\cdot \mathbb{E}_t\left[\sum_{\tau=1}^{\infty}(\beta)^{\tau}\left(\beta v(\gamma_{j,t+\tau})\right)^{\nu/(\nu-1)}\left(\frac{\varphi_{j,t+\tau}}{\nu}\right)^{1/(\nu-1)}\middle|\gamma_{jt},\varphi_{jt}\right]. \end{aligned} \tag{27}$$

The first term on the right hand side of (27) is firm value from past innovations, while the second term is the value generated by today's innovation (given today's γ and φ). Finally, the last term is the expected value generated from future innovations, which depends on today's γ and φ as long as there is some serial correlation in these states. Hence, a firm's valuation consists of both a forward looking and a backward looking part. A firm might have a high valuation today because of a series of good γ and φ draws in the past or because it is expected to have a good R&D technology in the future (given today's states). As a consequence, cross-sectional variation in Tobin's Q may come from past, present or expected future R&D activities.

In contrast to Tobin's Q, the ratio of the ex-dividend stock price to earnings gross of

R&D along the BGP is

$$
\begin{aligned}
\frac{P\left(\{\omega_{j\iota,t}\}_{\iota}, n_{j,t}, \gamma_{j,t}, \varphi_{j,t}\right)}{\tilde{E}(\{\omega_{j\iota,t}\}_{\iota}, n_{j,t})} &= \frac{V(\cdot) - \tilde{E}(\cdot) + \frac{1}{\varphi_{j,t}} x(\gamma_{j,t}, \varphi_{j,t})^{\nu}}{\tilde{E}(\cdot)} \\
&= \frac{\beta(1-\bar{x})}{1-\beta(1-\bar{x})} \\
&+ \frac{\left(\beta v(\gamma_{j,t})\right)^{\nu/(\nu-1)} \left(\frac{\varphi_{j,t}}{\nu}\right)^{1/(\nu-1)}}{\tilde{E}(\{\omega_{j\iota,t}\}_{\iota}, n_{j,t})} \\
&+ \left(1 - \frac{1}{\nu}\right) \cdot \mathbb{E}_t \left[\left. \frac{\sum_{\tau=1}^{\infty} \beta^{\tau} \left(\beta v(\gamma_{j,t+\tau})\right)^{\nu/(\nu-1)} \left(\frac{\varphi_{j,t+\tau}}{\nu}\right)^{1/(\nu-1)}}{\tilde{E}(\{\omega_{j\iota,t}\}_{\iota}, n_{j,t})} \right| \gamma_{j,t}, \varphi_{j,t} \right].
\end{aligned} \tag{28}
$$

The first term reflects past innovations. It is common across firms because all firms face the same rate of creative destruction. The second term depends on the current step size and R&D efficiency of a firm. Firms with higher step sizes and R&D efficiency innovate more and add value in the next period, raising the value of the firm relative to its current earnings. The third term captures the expected contribution of future innovations and depends on the stochastic process for the step size and R&D efficiency.

Finally, the model makes sharp predictions about the dynamics of a firm's sales and profitability. The share of sales of a firm relative to its total output, S, is equal to the number of its lines, i.e.,

$$
S_{jt} = n_{jt}. \tag{29}
$$

Sales of an individual firm fluctuate along the BGP as it is hit by shocks and adds and loses product lines. As each firm controls a *measure* of product lines n_{jt} in its portfolio, a firm never loses all its leading market positions due to creative destruction and there is no exit and entry in the model. The size of a firm is fully determined by its history of all its φ and γ realizations. Profitability, measured by earnings gross of R&D spending relative to sales, is given by

$$
\frac{\tilde{E}(\{\omega_{j\iota,t}\}_{\iota}, n_{jt})}{S_{jt}} = \sum_{\iota} \omega_{j\iota,t} \pi(\gamma_{\iota}). \tag{30}
$$

This object likewise depends on the history of past innovation activities (and step sizes in particular).

Model mechanisms The model generates dispersion in P/E ratios, expected growth, and R&D intensity across firms. It gives rise to rich dynamics and co-movement for firm markups, sales, R&D expenditure, and earnings.

In Klette and Kortum (2004) the expected innovation rate of a firm along a balanced growth path is unrelated to its size. This is because the Klette-Kortum R&D cost function is scaled by the number of lines a firm is active in. In (6) we do no such scaling. As a consequence our model can potentially generate a deviation from Gibrat's Law. At the same time, as illustrated below, with fully persistent φ and γ differences, our model can still generate a case in which the expected (sales) growth rate is unrelated to size.

In our model, variables like firm-level sales, earnings gross of R&D, and markups depend on the history of φ and γ states. High step size and high R&D efficiency firms innovated more in the past, grew faster, and added more profitable product lines. Other objects, like today's R&D expenditures, how much a firm's sales grows, and how profitable these newly added lines are only depends on today's φ and γ. A firm's expected growth contribution depends on its expected future realizations of its φ and γ.

To build intuition it is helpful to consider two extreme cases. First, suppose that φ and γ are fully persistent. In this case, firm sales are constant along a BGP and equal to $n^\star(\gamma_j, \varphi_j) = \left(\beta v(\gamma_j)\frac{\varphi_j}{\nu}\right)^{1/(\nu-1)} / \bar{x}^\star$. In this case a firm's value, R&D, earnings, and sales are all proportional to $v(\gamma_j)^{\nu/(\nu-1)}\varphi_j^{1/(\nu-1)}$. Hence, Gibrat's Law holds and firm sales growth is unrelated to firm sales. With just heterogeneity in φ a firm's contribution to growth is likewise proportional to its size.[11] Of course, such a case of fully persistent φ and γ does not generate any differences in firm price-earnings ratios as there is no dispersion in expected earnings growth. This case would also generate no relationship between R&D intensity and firm sales (as documented in Figure 4 above).

At another extreme, suppose that the φ and γ shocks are iid. In this case objects like sales or profitability that depend on the history of shocks are unrelated to forward looking components of the firm's valuation. In this case the model predicts a very negative relationship between sales growth or R&D intensity and size. With iid shocks,

[11] With heterogeneity in γ, firms with high step sizes contribute more to growth than suggested by their size. In this case a firm's growth contribution is given by its innovation activity x, which in this case is proportional to its size, times the logarithm of its step size (see (19)). Hence, the model could even generate a case wherein larger firms contribute more than proportionally to growth.

all firms—irrespective of their size—are expected to contribute the same to future growth in absolute terms.

As these polar cases illustrate, the persistence of shocks shapes key model predictions. The correlation between γ and φ also has important implications. If φ and γ are positively correlated with each other, then big firms have many product lines with large step sizes and therefore exhibit high (average) markups. If the covariance between φ and γ is sufficiently negative, however, then larger firms have lower (average) markups, and sales and markups comove negatively (the profitability of high sales growth firms is shrinking).

4 Calibration and consequences

In this section, we calibrate the model and calculate the growth contribution of firms by size. For parsimony, we set the Markov chain for shocks to approximate a VAR(1) process using the method described in Gospodinov and Lkhagvasuren (2014). Namely, we calibrate parameters of the process

$$\begin{pmatrix} \log(\gamma_t - 1) \\ \log\varphi_t \end{pmatrix} = \begin{pmatrix} \rho_\gamma \log(\gamma_{t-1} - 1) \\ \rho_\varphi \log\varphi_{t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{\gamma,t} \\ \varepsilon_{\varphi,t} \end{pmatrix}$$

$$\begin{pmatrix} \varepsilon_{\gamma,t} \\ \varepsilon_{\varphi,t} \end{pmatrix} \overset{iid}{\sim} N\left(\begin{pmatrix} \mu_\gamma \\ \mu_\varphi \end{pmatrix}, \begin{bmatrix} \sigma_\gamma^2 & \sigma_{\gamma,\varphi} \\ \sigma_{\gamma,\varphi} & \sigma_\varphi^2 \end{bmatrix} \right).$$

We use this method to generate $N = 625$ discrete combinations of γ and φ (25 distinct step sizes and R&D efficiencies) and the associated transition matrix for the Markov process of step size and R&D efficiency.[12]

We calibrate 9 parameters to target 10 moments. The parameters are the curvature of the R&D cost function (ν), the drift (μ_γ, μ_φ), persistence ($\rho_\gamma, \rho_\varphi$) and volatility ($\sigma_\gamma, \sigma_\varphi, \sigma_{\gamma,\varphi}$) of the step size and R&D efficiency VAR(1) process, and the discount factor

[12]We choose the number of grid points until the model solution does not change significantly when we increase the number further.

β. The ten moments are the growth rate of aggregate earnings net of labor input growth and growth in the number of firms in the sample, the average firm-specific discount rate from David et al. (2022), the contribution of the earnings-to-sales ratio to the relationship between the average 5-years earnings growth and initial $\log P/(E + R\&D)$, the contribution of earnings gross of R&D to sales ratio to the relationship between 5 year earnings-to-sales growth and $\log P/(E + R\&D)$, the aggregate earnings to aggregate sales ratio, the dispersion of predicted average 5 year growth in earnings and pre-R&D earnings in Table 1, the relationship between R&D intensity and sales in Figure 4, 90/50th percentile sales dispersion, and the dispersion of the ratio of price to earnings before R&D. The text adjacent to Table 1 details how we calculate the contributions to the relationship between earnings growth and P/E ratio.

The model is non-linear and all nine parameters jointly determine the moments. That said, some moments are more influential for some parameters. The aggregate earnings/sales and growth in aggregate earnings together are informative about the mean step size and R&D efficiency parameters, μ_γ and μ_φ. Given the aggregate growth moment, the discount rate pins down β. The dispersion in pre-R&D earnings growth helps pin down the step size volatility parameter σ_γ, while the dispersion of sales is informative about the volatility of R&D efficiency governed by σ_φ. The growth of net earnings is less dispersed than the growth of gross earnings, indicating that high step draws incur higher R&D cost and hence a negative covariance between φ and γ.

The contribution of earnings/sales is informative of the R&D cost curvature parameter ν. When curvature is higher, firm R&D choices, and hence the number of products they innovate upon, respond less to the step size and R&D efficiency draws. As a result, a firm's earnings growth comes less from variation in the number of product lines it adds and more from growth in earnings/sales. The contribution of the pre-R&D earnings-to-sales ratio is informative of the persistence of step size draws ρ_γ. The R&D intensity relationship with sales is informative of the persistence of R&D efficiency ρ_φ. All else equal, higher ρ_φ generates more dispersion in sales and a flatter relationship between R&D/sales with sales. The dispersion of price to earnings additionally disciplines the volatility of step size and R&D efficiency shocks.

Table 2: Calibration target moments

Targets	Data	Model
1. Growth in aggregate earnings, net of labor input growth, $g^\star$	1.2%	1.1%
2. Average firm-specific discount rate, $r^\star$	9.7%	9.6%
3. Contribution of $g_{E/S}$ to g_E on log $P/(E+R\&D)$	43%	43%
4. Contribution of $g_{(E+R\&D)/S}$ to $g_{E/S}$ on log $P/(E+R\&D)$	96%	99%
5. Aggregate earnings/aggregate sales	0.12	0.12
6. 90–10 range of fitted g_E	0.09	0.09
7. 90–10 range of fitted $g_{E+R\&D}$	0.15	0.15
8. R&D/sales projected on log sales	-0.028	-0.029
9. 90:50 sales ratio	17.5	17.1
10. Standard deviation of log $P/(E+R\&D)$	0.64	0.67

Notes: 1. earnings from I/B/E/S and labor input growth from the BLS. 2. average of firm-specific discount rate from David et al. (2022). 3. to 10. are manufacturing firms from CRSP, I/B/E/S, and Compustat.

Table 2 displays the target data moments and our model's fit to them.[13] Table 3 displays the resulting calibrated parameter values. The implied cost weighted price-cost markup is 1.14, similar to the estimate in Edmond, Midrigan and Xu (2023) for Compustat firms. We find that the step size is quite persistent with $\rho_\gamma = 0.94$ while the R&D efficiency has a persistence of $\rho_\varphi = 0.83$. Shocks to step sizes and R&D efficiency have a correlation of -0.59, which means that firms drawing a higher step size are likely to also draw a high R&D cost. Thus making big innovations typically requires more R&D spending. The model needs a high variance of R&D efficiency to generate the wide sales dispersion seen in the data. Step size heterogeneity is less powerful in generating big firms because the profit share for a product asymptotes to one as the step size rises, and because of the steep curvature in R&D costs ($\nu = 3.9$).

[13] Because we use R&D data to calculate some of the moments, we restrict the sample to manufacturing firms. In the sample of all firms, we find a stronger contribution (78% vs. 43%) of $g_{E/S}$ to the predictive power of the P/E ratio for earnings growth, larger dispersion in the predicted growth rate of net earnings (0.18 vs. 0.09), and a lower 90/50 ratio of sales (14.8 vs. 17.5) than when we restrict the sample to manufacturing firms.

Table 3: Calibrated parameter values

Parameter		Value
Curvature of R&D cost function	ν	3.91
Persistence of γ process	ρ_γ	0.94
Mean of shocks to γ process	μ_γ	-0.05
Variance of shocks to γ process	σ^2_γ	0.76
Persistence of φ process	ρ_φ	0.83
Mean of shocks to φ process	μ_φ	-4.79
Variance of shocks to φ process	σ^2_φ	49.2
Correlation between shocks	$\frac{\sigma_{\gamma\varphi}}{\sigma_\gamma\sigma_\varphi}$	-0.59
Discount factor	β	0.92

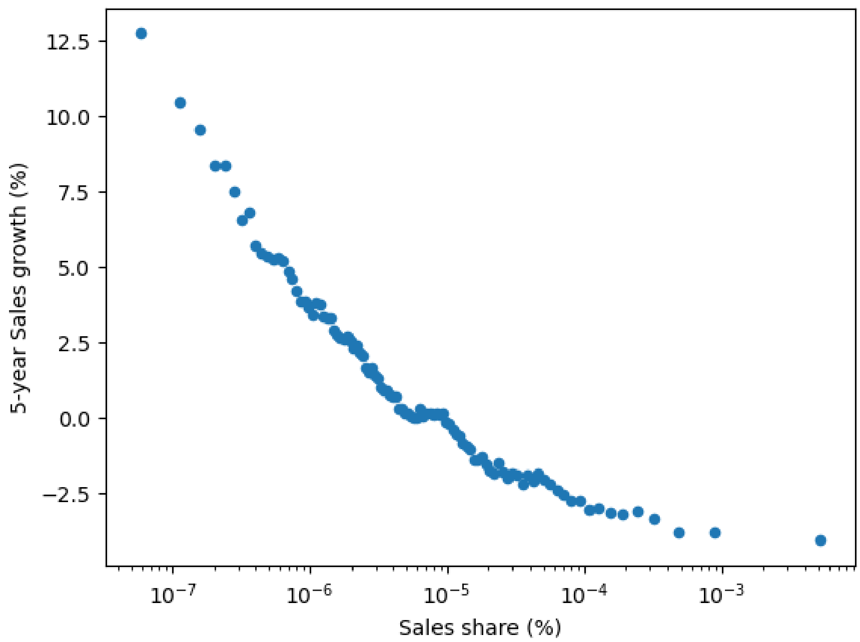
4.1 Untargeted moments

Our model is consistent with several empirical patterns that are of interest to the growth literature even though we do not target these moments in our calibration. Figure 6 displays the relationship between sales growth and sales in the calibrated model. Sales growth declines with sales as in Figure 5.

Interestingly, while Gibrat's Law fails in our model (as in the Compustat data), our calibrated model nevertheless generates a Pareto tail. The blue line in Figure 7 plots the complementary cumulative distribution function (CCDF) against sales share in the model while the orange line is the relationship according to a Pareto distribution with a tail parameter of 0.97. This is close to the 1.19 that we calculate in Compustat data.[14] Thus, the calibrated model is consistent with the empirical literature that documents a Pareto right tail of the firm size distribution, such as Axtell (2001). Whereas the literature that relies on random growth emphasizes that Gibrat's Law combined with an exit threshold is *sufficient* to generate a Pareto distribution, our model illustrates that Gibrat's Law is not *necessary* for doing so. Bursts in innovation potential due to stochastically high R&D efficiency and high step sizes are able to populate a Pareto tail

[14]We estimate the tail parameter as the negative of the OLS coefficient of the log CCDF of sales on log sales using logarithmically spaced size bins over the top decile of firms.

Figure 6: Sales growth vs. sales, baseline model

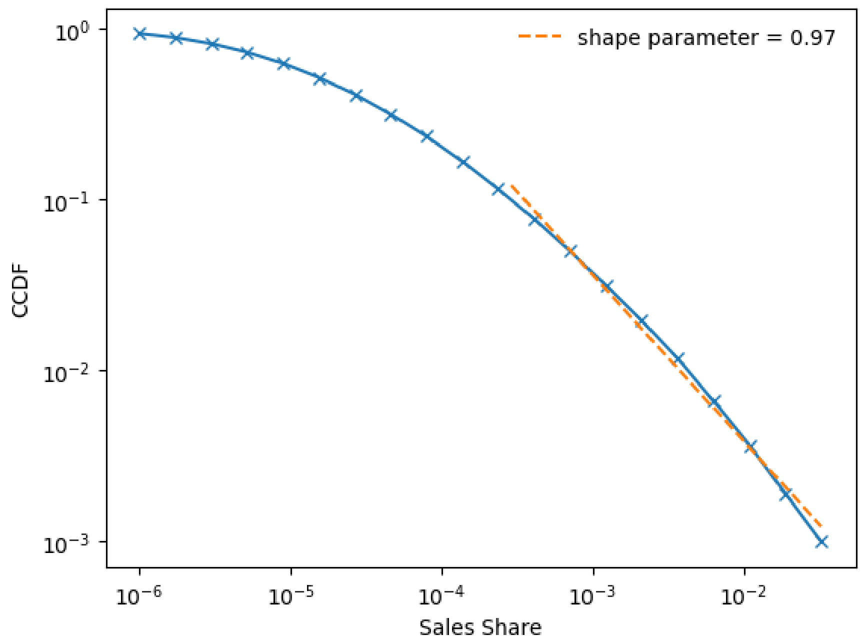


Notes: Model simulation. The figure is a binned scatterplot with 100 bins. The x-axis is sales relative to total sales across firms in the simulated data. The y-axis is average yearly growth of sales from year t to year $t+5$.

of firms. This is in the same spirit as the product innovation bursts documented by Berlingieri et al. (2025) in French manufacturing, which they show are crucial for generating a Pareto right tail of the firm size distribution. Our R&D efficiency and step size shocks generate such innovation bursts endogenously in our model.

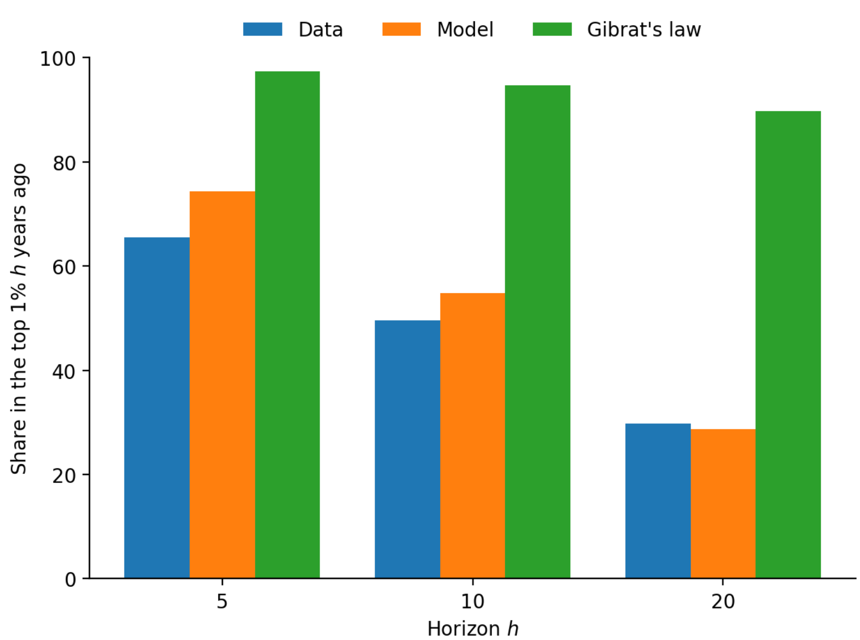
The model also reproduces the amount of churn observed in the right tail of the size distribution. Figure 8 presents one measure of persistence in this upper tail. It reports, for $h = 5, 10, 20$, the fraction of firms that are in the top 1% of the sales distribution in year t and were also in the top 1% in year $t-h$. The blue bars correspond to the data, and the orange bars to the calibrated model. At each horizon, the model roughly matches the empirical persistence. The green bars show the implications of a model that is in line with Gibrat's Law. We compute this by increasing the persistence of φ such that firm sales growth becomes independent of current sales. This Gibrat's Law benchmark generates markedly higher persistence of top 1% firms than observed in the data over all horizons.

Figure 7: Pareto tail in the sales share distribution, baseline model



Notes: Pareto tail coefficient estimated using the top 10% of the sales share distribution.

Figure 8: Persistence of top firms



Notes: Each bar reports the share of firms in the top 1% of sales that were also in the top 1% h years earlier, for horizons $h = 5, 10, 20$. The "Gibrat's law" bars simulate the model under an alternative parameterization with $\rho_\varphi = 0.9$ (versus 0.83 in the baseline) which approximately generates sales growth being independent of firm size.

In the data, firm sales growth covaries negatively with earnings/sales growth conditional on a firm's P/E ratio. The calibrated model replicates this pattern even though we do not target it. In the model, the correlation of $\Delta \log S_{it}$ and $\Delta \log(E_{it}/S_{it})$ is -0.32. In the model, sales growth comes from either larger step sizes or higher R&D efficiency. Margin (earnings/sales) growth occurs when a drawn step size exceeds the portfolio's average step size. In the calibrated model, firms with faster sales growth are more likely to draw higher R&D efficiency, but their step sizes are typically small because R&D efficiency shocks are negatively correlated with step-size shocks. As a result, firms that grow faster in sales show weaker margin growth.

4.2 Growth contribution of a group of firms

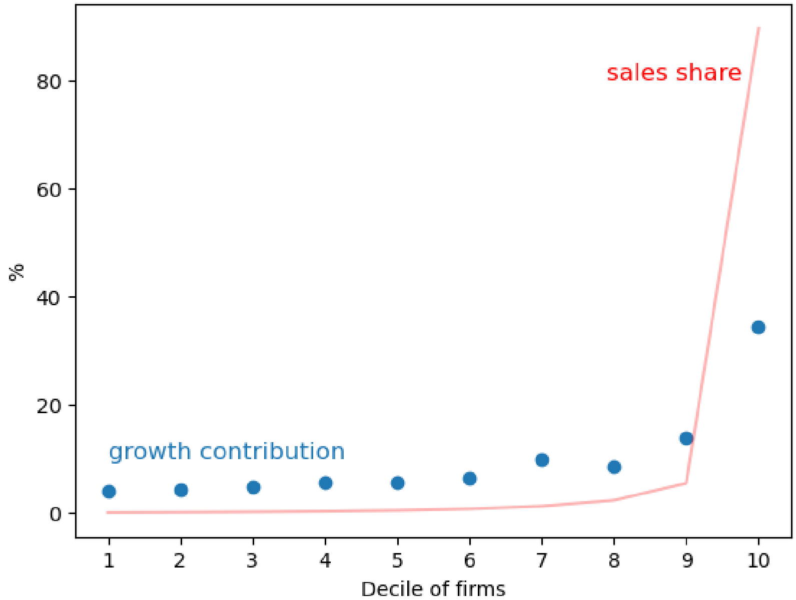
We now use the calibrated model to calculate firm contributions to growth. For a group of firms I, its contribution to aggregate growth over the next year is

$$m(I) \cdot \frac{\sum_k s_k^I \cdot x(\gamma_k, \varphi_k) \cdot \log(\gamma_k)}{\sum_k s_k \cdot x(\gamma_k, \varphi_k) \cdot \log(\gamma_k)} \tag{31}$$

where s_k^I is the share of firms with step size γ_k and R&D efficiency φ_k in the group I. $m(I)$ is the fraction of firms in I. A special case is when step sizes and R&D efficiency shocks are iid. In this case, the contribution is 1% for all size percentiles because $s_k^I = s_k$. Another extreme case is when a firm's growth contribution is proportional to its sales share as in Klette and Kortum (2004).

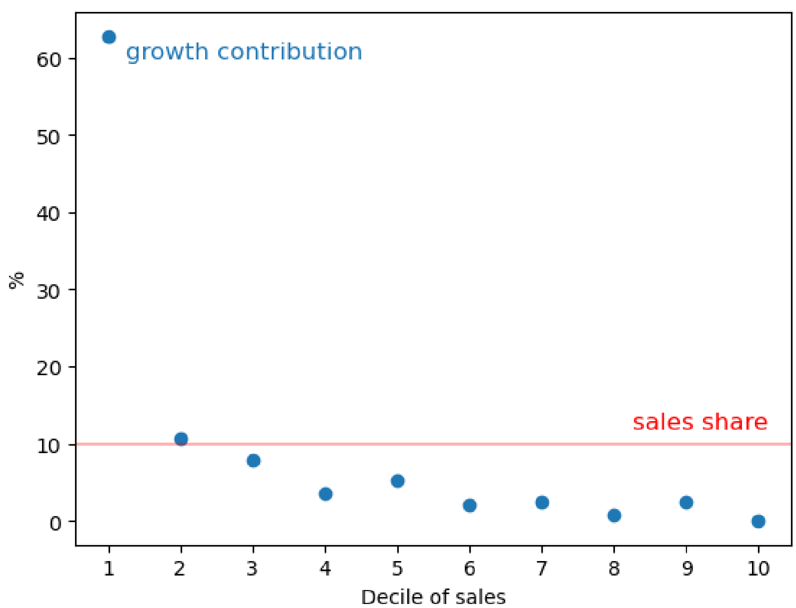
Applying (31) in our calibrated model, we find that large firms do contribute more to growth but less than proportionally to their sales share. Figure 9 ranks firms by sales and shows the contribution of firms in each size decile. The largest 10% of firms account for less than 40% of aggregate growth (blue dots), whereas their sales share is over 80% (red line). Figure 10 shows this more clearly by plotting the growth contribution by sales deciles. The largest firms that account for 10% of sales contribute much less than 10% to aggregate growth. Meanwhile, the smallest firms that account for 10% of sales are responsible for over 60% of aggregate growth.

Figure 9: Contribution to growth and sales share by firm decile



Notes: Model simulation. The figure shows the contribution to growth and sales share of firms in each firm decile.

Figure 10: Contribution to growth and sales share by sales decile



Notes: Model simulation. The figure shows the contribution to growth and sales share of firms in each sales decile.

4.3 Idiosyncratic shocks, growth, and allocative efficiency

Idiosyncratic shocks to the innovation step size and R&D efficiency are a key feature of the model for fitting the relationship between earnings growth and the P/E ratio. To quantitatively assess the importance of the two shocks we run two counterfactuals. First, we calculate the counterfactual growth rate and allocative efficiency when we give all firms the same step size conditional on their R&D efficiency φ:

$$\gamma^c(\varphi) = \exp(\mathbb{E}[\log(\gamma - 1)|\log\varphi]) + 1.$$

This removes step size heterogeneity conditional on φ but preserves the covariance between step size and R&D efficiency across firms.[15] We keep all the other parameters the same, with the stochastic process for R&D efficiency having the same drift, persistence, and variance as in the baseline calibration.[16] To isolate the effect of removing γ heterogeneity on growth, we implement a subsidy τ on R&D to keep the aggregate R&D to aggregate output ratio the same as in the baseline. In this counterfactual the firm's R&D decision x^c solves

$$\max_{x^c \geq 0} \left\{ \beta x^c v(\gamma^c(\varphi)) - (1-\tau)\frac{1}{\varphi}(x^c)^\nu \right\}. \tag{32}$$

We set the R&D subsidy rate τ such that total R&D relative to output is the same in both economies:[17]

$$\sum_{k=1}^{N} s_k \frac{1}{\varphi_k}\left((x_k^c)^\nu - x_k^\nu\right) = 0. \tag{33}$$

[15]This counterfactual removes most of the dispersion in step sizes. The dispersion in step size across firms can be decomposed into dispersion conditional on φ and dispersion across φ:

$$\underbrace{Var(\log(\gamma-1))}_{6.85} = \underbrace{\mathbb{E}[Var(\log(\gamma-1)|\log\varphi)]}_{5.08} + \underbrace{Var(\mathbb{E}[\log(\gamma-1)|\log\varphi])}_{1.77}$$

Almost 75% (5.08/6.85) of the dispersion in step size comes from dispersion conditional on R&D efficiency. The covariance of step size and R&D efficiency $Cov(\log(\gamma-1), \log\varphi)$ is -16.94 in both the baseline and the counterfactual.

[16]We calculate the unweighted average of step sizes across firms with the same φ in the ergodic distribution. We re-solve the model with each firm's R&D efficiency following the calibrated AR(1) process $\log\varphi_t = \rho_\varphi \log\varphi_{t-1} + \epsilon_{\varphi,t}$ and $\epsilon_{\varphi,t} \overset{iid}{\sim} N(\mu_\varphi, \sigma_\varphi^2)$, setting γ_{jt} to $\gamma^c(\varphi_{jt})$.

[17]The R&D subsidy is 91.1% and 99.1% in the first and second counterfactual, respectively.

As a second counterfactual we instead remove the heterogeneity in R&D efficiency conditional on the step size. For firms with the same step size, we set their R&D efficiency to

$$\varphi^c(\gamma) = \exp(\mathbb{E}[\log \varphi | \log(\gamma - 1)]),$$

and again subsidize R&D such that aggregate R&D spending relative to output is the same as in the baseline calibration.

In addition to the growth rate, we are also interested in what the counterfactuals imply for allocative efficiency, which is the ratio of the geometric to the arithmetic average of the inverse markups across lines:

$$\text{Allocative efficiency } = \frac{\exp\left(\int_0^1 \log \frac{1}{\mu(i)}\, di\right)}{\int_0^1 \frac{1}{\mu(i)}\, di} \leq 1. \tag{34}$$

Allocative efficiency is less than one if there is markup dispersion across lines.

Table 4 shows the results of the two counterfactuals. The first row provides the baseline growth rate and allocative efficiency for comparison. The second row is the counterfactual that removes step size heterogeneity conditional on R&D efficiency φ. In this first counterfactual, the growth rate falls from 1.1% to 0.08%. This is because in the baseline high step size firms spend more on R&D conditional on φ, raising the R&D-weighted step size.

Misallocation due to markup dispersion lowers the level of output by 5% in the baseline model but becomes negligible in the first counterfactual. This is because, in the baseline calibration, most of the step size dispersion is conditional on φ, which is removed in the counterfactual.

The third row of Table 4 shows the second counterfactual that removes R&D efficiency heterogeneity conditional on the step size. In this case, the growth rate collapses to 0.005% while allocative efficiency improves from 0.950 to 0.978. Similar to the previous counterfactual, growth declines because the counterfactual removes firms with high step size and high R&D efficiency. Markup dispersion does not completely disappear because we maintained the dispersion in step sizes. Taken together these two counterfactual exercises highlight the importance of firms with both high step size and high R&D efficiency for aggregate growth.

Table 4: Effect of removing step size or R&D efficiency heterogeneity

	Growth rate	Allocative efficiency
Baseline	1.1%	0.950
$\gamma^c(\varphi) = \exp(\mathbb{E}[\log(\gamma - 1) \| \log \varphi]) + 1$	0.08%	1.000
$\varphi^c(\gamma) = \exp(\mathbb{E}[\log \varphi \| \log(\gamma - 1)])$	0.005%	0.978

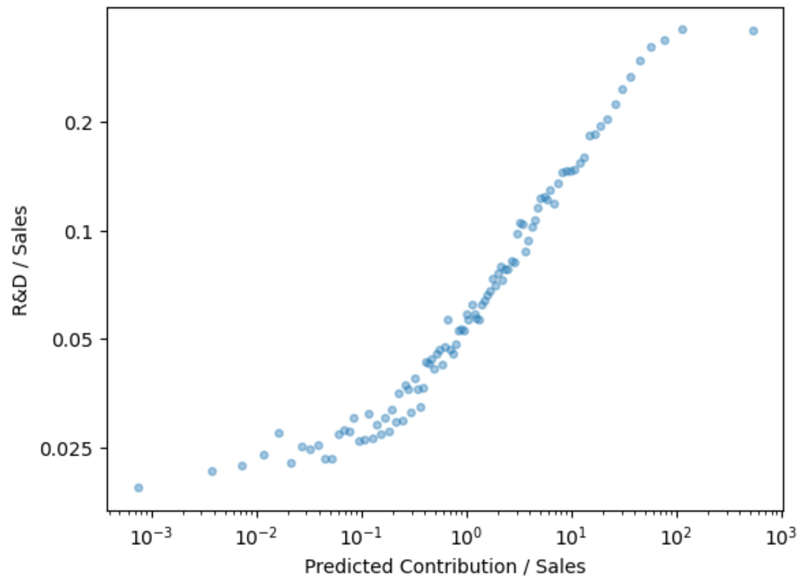
4.4 Empirical application: growth from individual firms

We can use our model to estimate the expected contribution of individual firms to future growth in the data. Although we do not observe the step size or R&D efficiency of a company's current innovations, we construct a proxy by estimating a nonlinear relationship in the calibrated model between firm observables and a firm's expected growth contribution over the next 5 years.[18] For firms with positive earnings, we estimate their expected growth contribution using their price–earnings ratio, earnings (net of any R&D) relative to sales, and sales. For firms with non-positive earnings, we estimate their expected growth contribution using their price–sales ratio (both in per share terms) and their sales.

In the model we estimate a firm's growth contribution to rise with its P/E ratio for firms with positive earnings, and with its P/S ratio for firms with negative earnings, controlling for firm sales and the earnings-to-sales ratio. This is because sales and earnings/sales control for past innovation step sizes and R&D efficiency. Conditional on these, higher P/E and P/S reflect better current and future innovation abilities, which contributes to aggregate growth. Furthermore, a firm's growth contribution increases with its sales share conditional on its P/E, P/S, and E/S ratios, and in its E/S ratio conditional on its current sales and P/E ratio. The intuition is the same for both findings: Fixing the P/E ratio is akin to fixing a firm's current and future innovation ability relative to its past. Since a high earnings firm had better draws in the past than a low earnings firm, the future innovation ability of a high earnings firm must be superior to that of a lower earnings firm with the same P/E ratio. Appendix Figure A18

[18] We estimate this nonlinear mapping using a generalized additive model. Each regressor enters through a cubic spline with four basis functions, and a smoothing parameter selected via cross-validation.

Figure 11: R&D/Sales vs. expected growth contribution/Sales



Notes: Data from CRSP, I/B/E/S, and Compustat. Manufacturing firms only. The figure is a binned scatterplot with 100 bins. The y-axis is R&D intensity across manufacturing firms in a year. The x-axis is estimated expected growth contribution relative to sales.

details how a firm's expected growth contribution loads on each of these predictors.

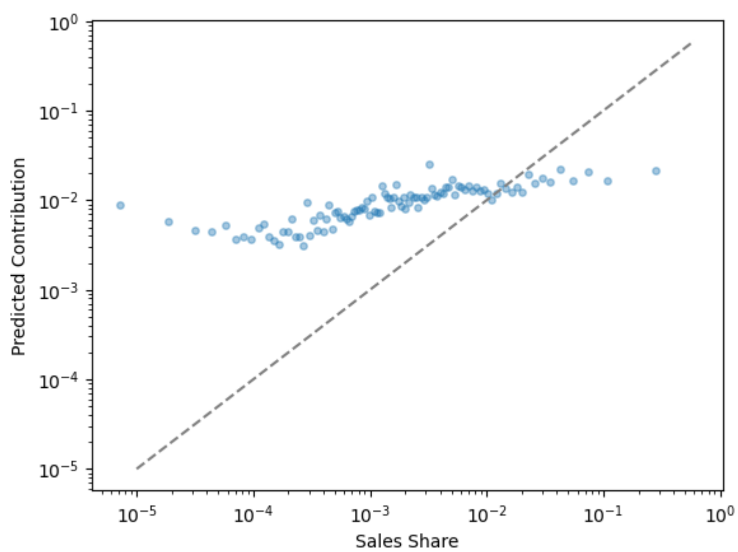
The proxy relationship fits the model simulated data fairly well, accounting for 71% of the variation in expected growth contributions in our model simulated data. In Appendix Table A7, we show the marginal R^2 of each regressor. Most of the explanatory power comes from the P/E and E/S ratios along with sales for positive earnings firms. As an additional check, Appendix Figure A19 compares the actual firm growth contribution in the model to the growth contribution implied by the proxy relationship by firm sales deciles. Both the model and fitted growth contributions decline with sales decile, so that smaller firms contribute more to growth than their share of sales.

We next apply the fitted relationship to actual firms in the Compustat sample to infer their expected growth contribution through the prism of our model. For manufacturing firms in Compustat, Figure 11 illustrates a strong positive association between a firm's expected growth contribution from the proxy relationship (expressed relative to sales) and its R&D to sales ratio in the data. This provides some reassurance that our measure of a firm's growth contribution is indeed related to its innovation efforts (as measured by its R&D spending).

Looking across all Compustat firms, Figure 12 says growth contributions increase with firm sales shares, but less than one-for-one (as shown by the 45 degree line). Firm growth contributions are larger (smaller) than their sales share for small (large) firms, as shown in Figure 13. The smallest firms that account for 10% of sales are expected to contribute around 63% of growth.

We can also use the proxy relationship to estimate the growth contribution of individual firms. In Table 5, we do so for the Magnificent Seven, which saw significant increases in their market capitalization in recent years. These firms represented about one-quarter of total market value over 2020 to 2024, and have higher than average earnings-to-sales ratios. They are expected to account for 7.3% of growth (among Compustat firms) over the five years from 2024, slightly below their 8% share of sales.[19] Within the seven firms, Nvidia and Tesla have expected contributions that significantly exceed their sales share.

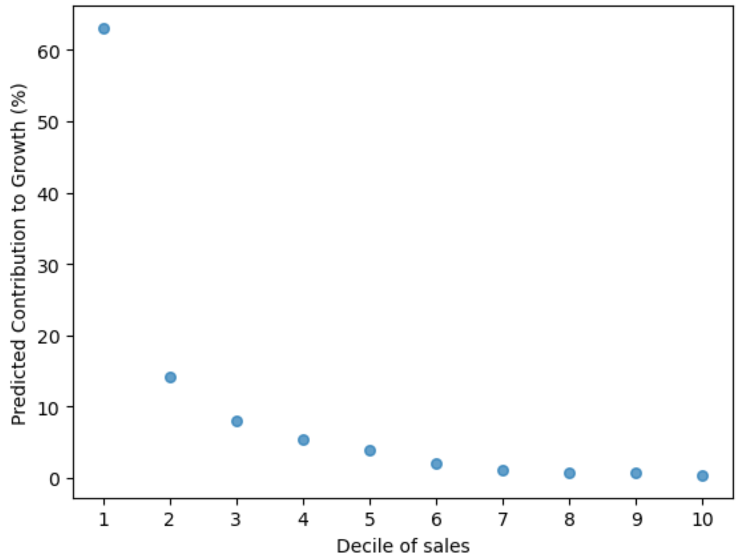
Figure 12: Expected growth contribution vs. sales share for Compustat firms



Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is sales relative to total sales across firms in a year. The y-axis is estimated expected growth contribution.

[19]As our sample of public firms accounts for approximately one-fourth of total private nonfarm employment, one can multiply our estimated growth contribution by one-fourth to infer the contribution to aggregate growth. This estimate may be conservative, however, if public firms are selected on their expected growth contribution conditional on their sales.

Figure 13: Predicted contribution to growth by sales decile for Compustat firms



Notes: Data from CRSP, I/B/E/S, and Compustat. The x-axis is sales decile in a year. The y-axis is estimated expected growth contribution.

Table 6 contrasts some of the Magnificent Seven firms with other large firms in their industry. Intel has a sales share that is more than double of Nvidia's but is expected to account for only 0.19% of all growth versus 3.86% for Nvidia. Walmart has a larger sales share than Amazon, but is expected to contribute much less to growth. General Motors and Ford are expected to contribute less to growth than Tesla despite having double the sales and comparable earnings to Tesla. These examples underscore that innovation giants of the past are not necessarily expected to be innovation leaders in the future.

Figure 14 shows the share of expected growth by the top 7 contributing firms versus the sales share of those same firms for each year from 1985 to 2024. Their sales share has been fairly stable, while their expected growth contribution is much higher than their sales share and varies more over time. Notice a local peak in the expected contribution of the top 7 firms in 2000. The expected contribution of the top 7 firms in recent years is much more modest. Table 7 displays the identity of the seven firms over time. The top contributors tend to be technology firms.

Table 5: Expected contribution of the Magnificent Seven, 2020–2024 average

Name	Share of Market Cap	Share of Earnings	Share of Sales	Exp. Growth Contribution	Contribution Rel. to Sales
Nvidia	1.34	0.63	0.14	3.86	26.8
Tesla	1.92	0.45	0.37	1.83	4.92
Amazon	3.71	1.45	2.74	1.24	0.45
Microsoft	5.20	3.62	1.02	0.23	0.22
Meta	1.70	2.10	0.66	0.06	0.10
Apple	5.86	4.87	1.93	0.05	0.03
Alphabet	3.53	3.74	1.45	0.03	0.02
Total	23.3	16.9	8.31	7.31	0.88

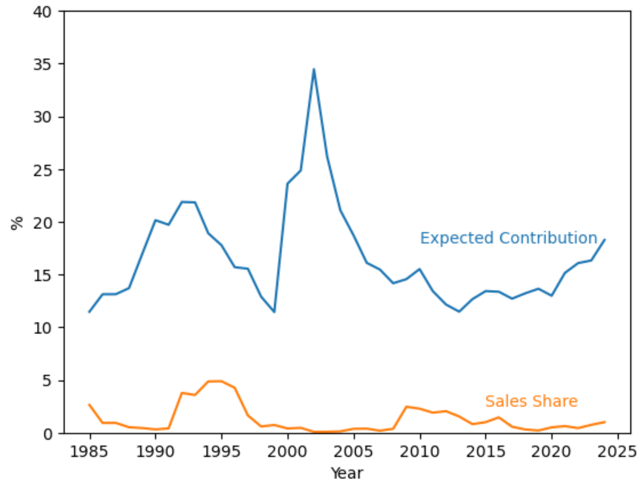
Notes: Data from CRSP, I/B/E/S, and Compustat. The first three columns show each firm's percent share of market capitalization, earnings, and sales relative to the total across all firms in the sample over 2020 to 2024. The fourth column estimates expected growth contribution for each firm using the proxy relationship described in the main text, where total expected growth contribution across all firms in the sample is normalized to 1. The final column shows a firm's expected contribution to growth relative to its sales share.

Table 6: Comparison of expected firm contributions, 2020–2024 average

Name	Share of Market Cap	Share of Earnings	Share of Sales	Exp. Growth Contribution	Contribution Rel. to Sales
Nvidia	1.34	0.63	0.14	3.86	26.8
AMD	0.38	0.21	0.10	0.75	7.33
Intel	0.40	0.70	0.36	0.19	0.53
Amazon	3.71	1.45	2.74	1.24	0.45
Walmart	0.91	0.92	3.11	0.0002	0.00005
Tesla	1.92	0.45	0.37	1.83	4.92
GM	0.14	0.55	0.81	0.00006	0.00007
Ford	0.12	0.32	0.83	0.00008	0.00009

Notes: See notes to Table 5.

Figure 14: Expected growth contribution of top 7 contributors over time



Notes: Data from CRSP, I/B/E/S, and Compustat. The blue line shows the combined expected contribution of the top 7 contributors in each year between 1985 and 2024. The orange line shows the combined sales share of these firms.

Table 7: Top 7 Firms by expected contribution by decade

	1990	2000	2010	2024
1	Lin Broadcasting	Yahoo! Inc	First Solar	Nvidia
2	Genentech	Veritas Software	American Tower	Intuitive
3	Newmont Gold Co	JDS Uniphase Crp	Las Vegas Sands	ServiceNow
4	Tejon Ranch	Microsoft	Citigroup	Mastercard
5	Alza	PeopleSoft	Bank of America	HubSpot
6	Cetus Corp	BroadVision Inc	Celgene	Tesla
7	Time Warner	Clear Channel	CME Group Inc	Cadence Design

Notes: Data from CRSP, I/B/E/S, and Compustat. Each column estimates expected growth contribution for each firm using the proxy relationship described in the text in the headline decade, where total expected growth contribution across all firms in the sample is normalized to 1.

5 Conclusion

This paper constructs an endogenous growth model with shocks to firm innovation step sizes and R&D efficiency, and calibrates it to match the relationship between P/E ratios, earnings, earnings growth, and earnings-to-sales growth across firms. The firm-specific shocks generate differences in future earnings and hence differences in valuation relative to current earnings. Our calibrated model implies that the largest (smallest) firms contribute much less (more) than their share of sales.

Our model suggests that a combination of market capitalization, earnings, and sales could be a good indicator of firms' contribution to growth. There are potential policy uses for real-time estimates of firm contributions to future productivity growth. Because innovations build on each other, large step-size firms confer larger knowledge spillovers. Aghion, Bergeaud, Boppart, Klenow and Li (2026) highlight potential growth gains from reallocating research effort from low to high step-size firms.

Our empirical analysis focused on publicly-listed firms. One could expand the sample to include non-listed firms for which there exist data on market value, such as from PitchBook. Ideally, one could incorporate moments on all firms in the economy regardless of whether they have market capitalization data. The fact that deviations from Gibrat's Law are more pronounced for publicly listed firms than for U.S. firms as a whole suggests that listed firms may be selected in part on their growth potential.

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Online Appendix (not for publication)

"Idea Rents and Firm Growth" by Boppart, Klenow, Laski, and Li

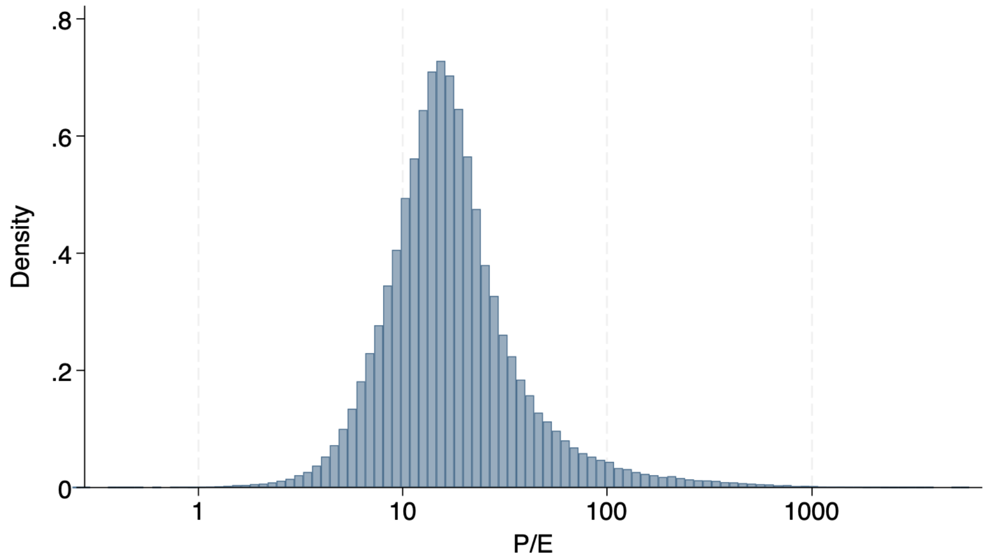
A Empirics

A.1 Significant dispersion in P/E ratios across firms.

Figure A1 shows the distribution of raw P/E ratios pooled across firm-years. Figure A2 shows the standard deviation and interquartile range across firms within each year of our sample. The degree of dispersion fluctuates noticeably, but is nontrivial in all years. Figures A3 and A4 show significant dispersion after taking out year fixed effects and fiscal-year end month effects or 4-digit SIC industry fixed effects.

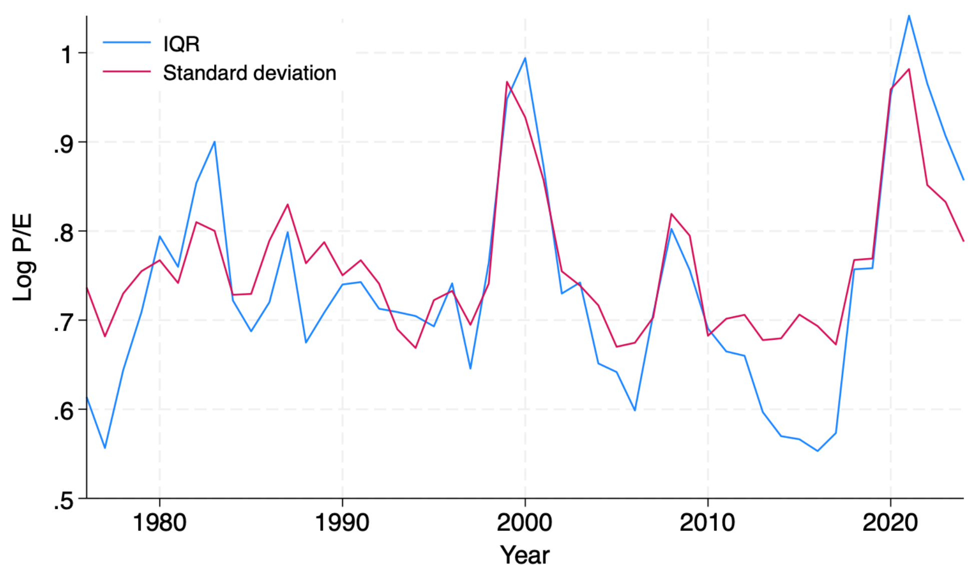
Differences in P/E ratios across firms could be due to firm-specific discount rates. However, the cross-sectional dispersion is not just about dispersion in firm-specific risks. Figure A5 plots the distribution of $\frac{P}{E}_{it}$ residualized on r_{it}, the firm-specific discount rate constructed following David, Schmid and Zeke (2022) using the Fama-French three-factor model. We find significant dispersion in the residual. This is consistent with the findings of Vuolteenaho (2002) that differences in firm-level stock returns are partly driven by changes in cash-flow expectations as opposed to entirely reflecting changes in the discount rate.

Figure A1: Distribution of P/E



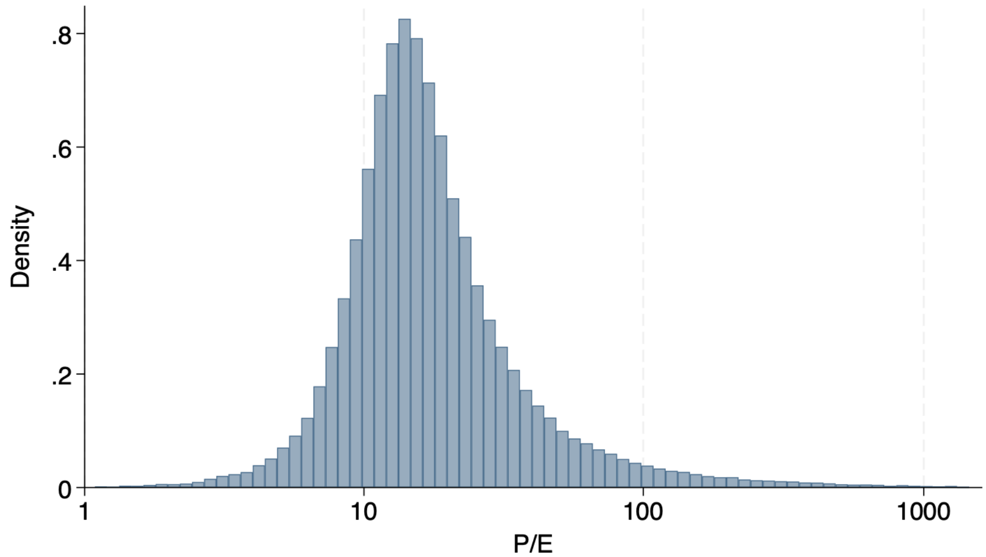
Notes: Data from CRSP and I/B/E/S. The figure plots the pooled distribution of the raw P/E ratio across firm-years. The top and bottom 0.1% of observations are trimmed.

Figure A2: Cross-firm dispersion of $\log P/E$ ratios over time



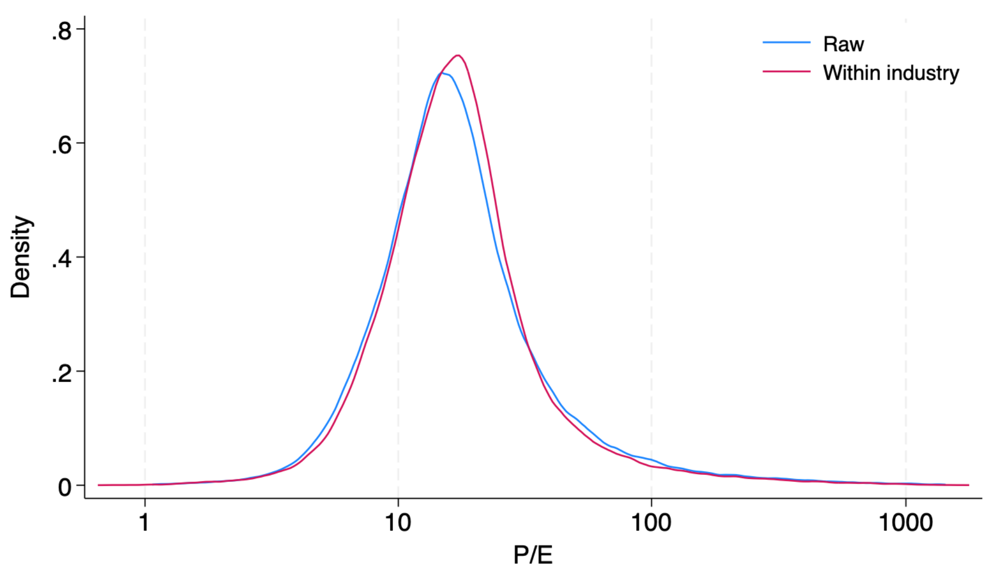
Notes: Data from CRSP and I/B/E/S. The lines plot the interquartile range and the standard deviation of $\log P/E$ within each year. The top and bottom 0.1% of observations are trimmed.

Figure A3: Distribution of P/E, removing year and fiscal month FE



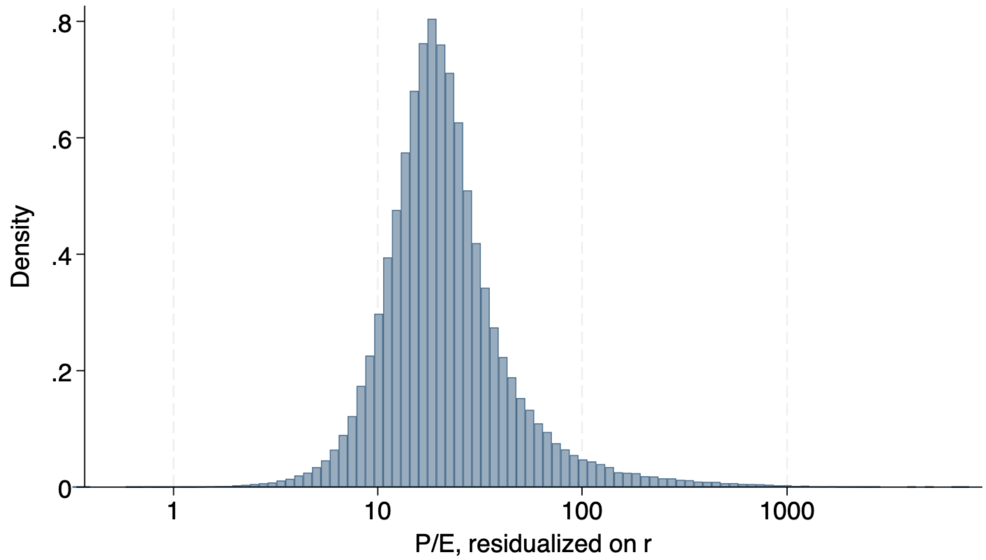
Notes: Data from CRSP and I/B/E/S. The figure plots the distribution of P/E after residualizing $\log P/E$ on year and fiscal-year end month fixed effects. The top and bottom 0.1% of observations are trimmed.

Figure A4: Distribution of P/E, removing industry FE



Notes: Data from CRSP, I/B/E/S, and Compustat. The figure shows the raw distribution of P/E and the distribution after residualizing $\log P/E$ on 4-digit SIC industry fixed effects. The top and bottom 0.1% of observations are trimmed.

Figure A5: Distribution of P/E, controlling firm-specific risk



Notes: Data from CRSP and I/B/E/S. The figure plots the distribution of P/E after residualizing $\log P/E$ on a cubic in the firm-specific discount rate r_{it} from David et al. (2022). The top and bottom 0.1% of observations are trimmed.

Table A1: Growth of earnings and earnings/sales in manufacturing

	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$
g_E on $\log(P/(E+R\&D))$	0.359	0.173	0.119	0.090	0.069
	(0.023)	(0.013)	(0.009)	(0.008)	(0.007)
$g_{E+R\&D}$ on $\log(P/(E+R\&D))$	0.230	0.135	0.099	0.077	0.068
	(0.007)	(0.005)	(0.004)	(0.003)	(0.003)
$g_{(E+R\&D)/S}$	0.144	0.074	0.049	0.034	0.028
	(0.006)	(0.004)	(0.003)	(0.003)	(0.002)
g_S	0.086	0.062	0.050	0.043	0.039
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)

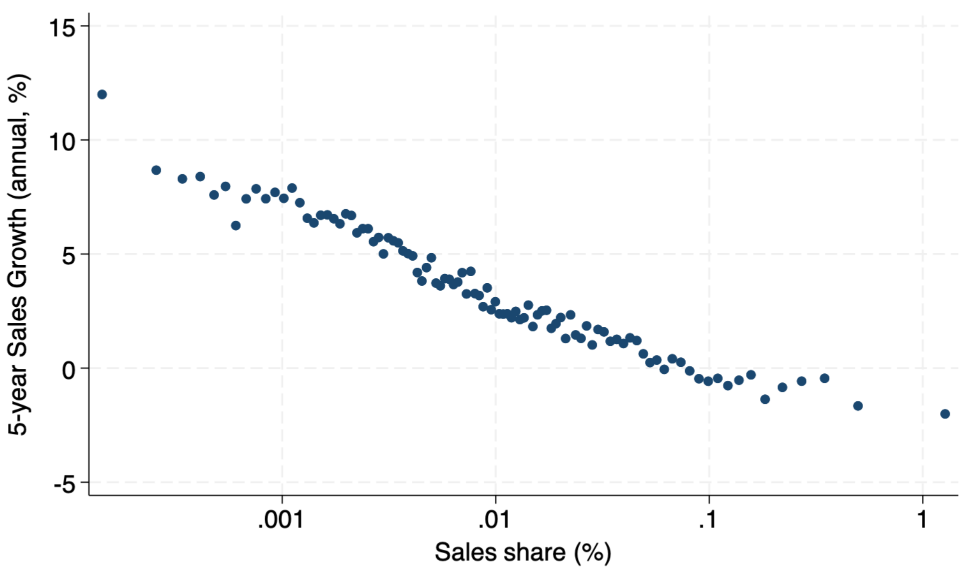
Notes: Data from CRSP, I/B/E/S, and Compustat. Manufacturing firms only. The first row displays the coefficient from regressing average yearly earnings growth from year t to $t+h$ on the $\log P/(E+R\&D)$ ratio in t. The second row uses growth in earnings gross of R&D. The last two rows decompose growth in earnings gross of R&D into growth in the ratio of earnings gross of R&D to sales and growth in sales. Each column corresponds to a horizon of h years, $h=1,\ldots,5$. All regressions control for year and fiscal-year end month fixed effects, as well as a cubic in the firm-specific discount rate from David et al. (2022).

Table A2: Growth of earnings and earnings/sales vs. P/E ratio, all industries

	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$
g_E on $\log(P/E)$	0.447	0.274	0.197	0.154	0.129
	(0.006)	(0.004)	(0.003)	(0.002)	(0.002)
$g_{E/S}$	0.384	0.228	0.159	0.122	0.101
	(0.006)	(0.004)	(0.003)	(0.002)	(0.002)
g_S	0.063	0.046	0.037	0.032	0.028
	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)

Notes: Data from CRSP, I/B/E/S, and Compustat. The first row displays the coefficient from regressing average yearly earnings growth from year t to $t+h$ on the $\log P/E$ ratio in t. The second and third rows decompose the coefficient into growth in the ratio of earnings to sales and sales growth. Each column corresponds to a horizon of h years, $h = 1, \ldots, 5$. All regressions control for year and fiscal-year end month fixed effects, as well as a cubic in the firm-specific discount rate from David et al. (2022).

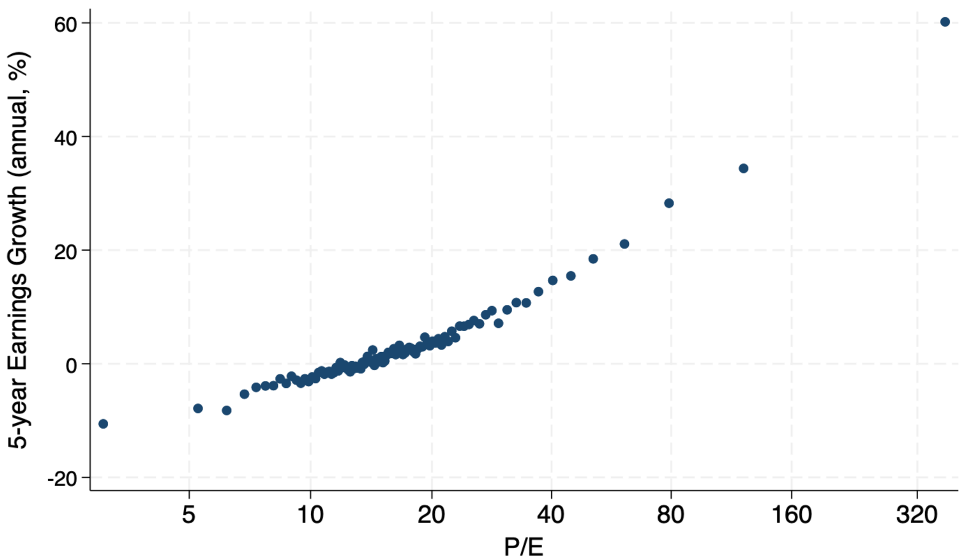
Figure A6: Sales growth vs. sales, all industries



Notes: Data from Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is sales relative to total sales across firms in t. The y-axis is average yearly growth of sales from year t to year $t+5$.

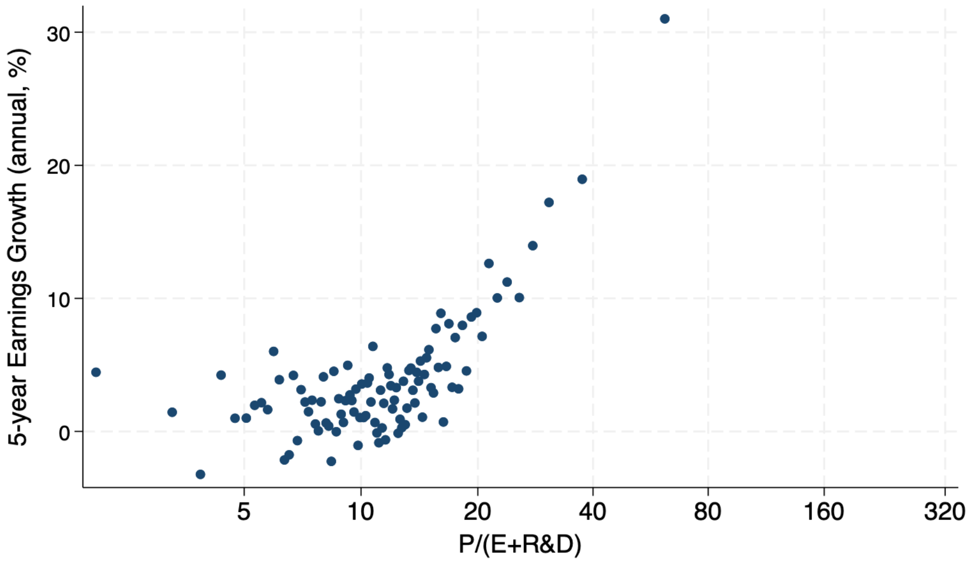
A.2 Motivating facts for firms with 10 or more years in the Compustat sample

Figure A7: Earnings growth vs. P/E for firms ≥ 10 years, all industry



Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the P/E ratio in year t. The y-axis is average yearly earnings growth from year t to $t + 5$. Both variables are residualized on a cubic of the firm-specific discount rate from David et al. (2022), fiscal-year end month and year fixed effects. Excludes each firm's first ten years in the sample.

Figure A8: Earnings growth vs. $P/(E + R\&D)$ for firms ≥ 10 years, manufacturing



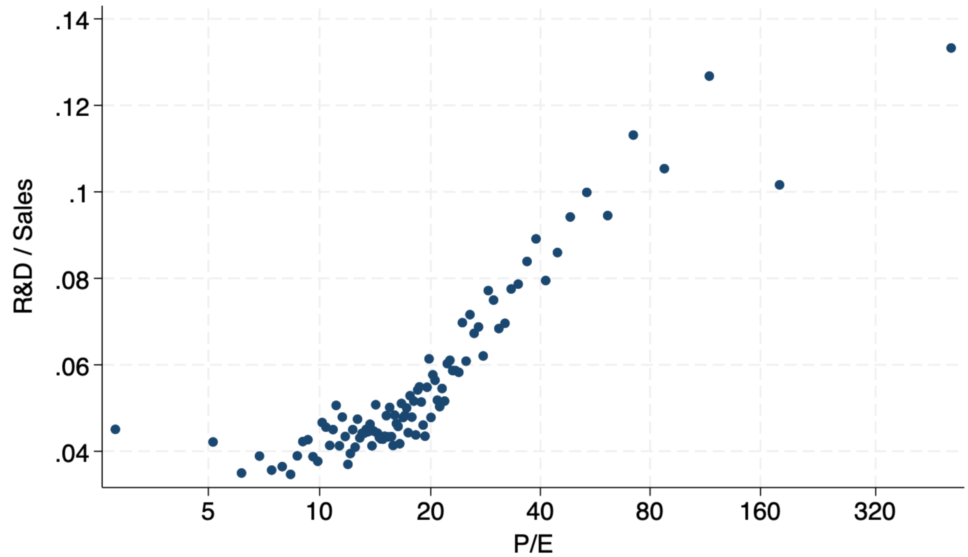
Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the $P/(E + R\&D)$ ratio in year t. The y-axis is average yearly earnings growth from year t to $t + 5$. Both variables are residualized on a cubic of the firm-specific discount rate from David et al. (2022), fiscal-year end month and year fixed effects. Manufacturing firms only. Excludes each firm's first ten years in the sample.

Table A3: Growth of earnings and earnings/sales in manufacturing for firms ≥ 10 years

	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$
g_E on $\log(P/(E+R\&D))$	0.349	0.165	0.110	0.081	0.064
	(0.024)	(0.014)	(0.010)	(0.008)	(0.007)
$g_{E+R\&D}$ on $\log(P/(E+R\&D))$	0.218	0.129	0.094	0.072	0.064
	(0.008)	(0.006)	(0.004)	(0.004)	(0.004)
$g_{(E+R\&D)/S}$	0.150	0.079	0.054	0.038	0.033
	(0.008)	(0.005)	(0.004)	(0.003)	(0.003)
g_S	0.069	0.050	0.039	0.034	0.031
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)

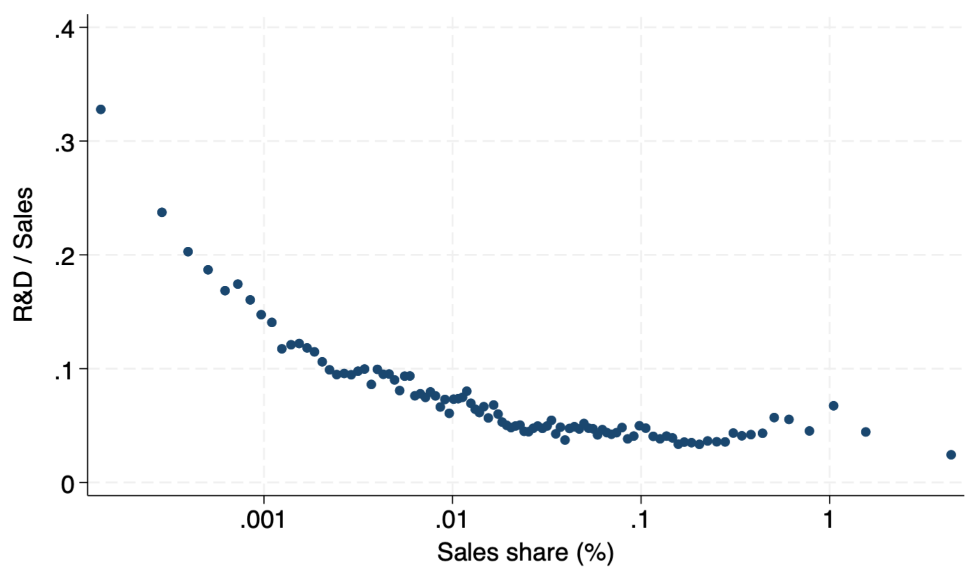
Notes: Data from CRSP, I/B/E/S, and Compustat. Manufacturing firms only. The first row displays the coefficient from regressing average yearly earnings growth from year t to $t+h$ on the $\log P/(E+R\&D)$ ratio in t. The second row uses growth in earnings gross of R&D. The last two rows decompose growth in earnings gross of R&D into growth in the ratio of earnings gross of R&D to sales and growth in sales. Each column corresponds to a horizon of h years, $h = 1, \ldots, 5$. All regressions control for year and fiscal-year end month fixed effects, as well as a cubic in the firm-specific discount rate from David et al. (2022). Excludes each firm's first ten years in the sample.

Figure A9: R&D/Sales vs. P/E for firms ≥ 10 years, manufacturing



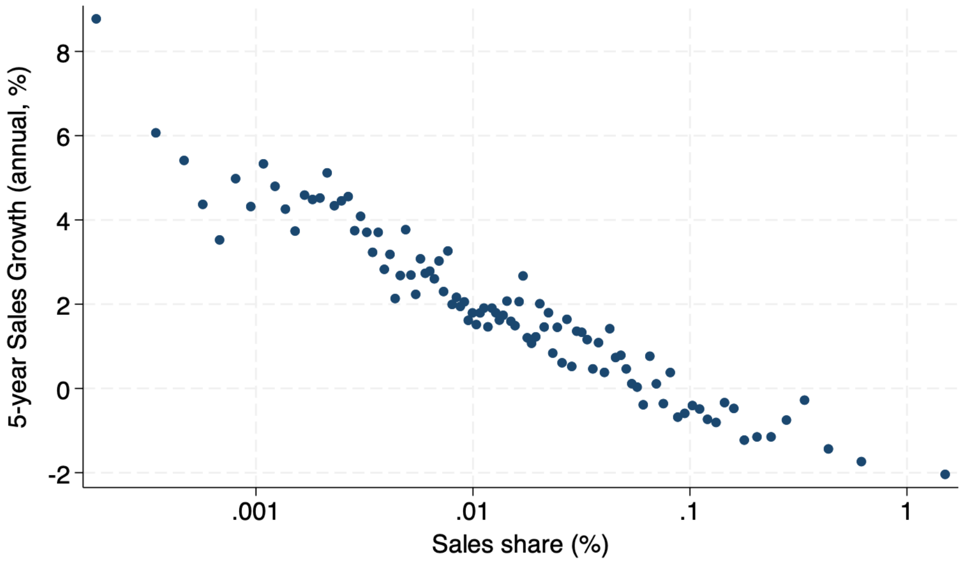
Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the P/E ratio in year t. The y-axis is R&D over sales in year t. Both variables are residualized on a cubic of the firm-specific discount rate from David et al. (2022), fiscal-year end month and year fixed effects. Manufacturing firms only. Excludes each firm's first ten years in the sample.

Figure A10: R&D/Sales vs. sales for firms ≥ 10 years, manufacturing



Notes: Data from Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is sales relative to total sales across firms in t. The y-axis is R&D over sales in year t. Manufacturing firms only. Excludes each firm's first ten years in the sample.

Figure A11: Sales growth vs. sales for firms ≥ 10 years, all industry



Notes: Data from Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is sales relative to total sales across firms in t. The y-axis is average yearly growth of sales from year t to year $t+5$. Excludes each firm's first ten years in the sample.

A.3 Adjusted earnings and market value

We adjust earnings by adding back interest payment to debt and subtracting user cost of capital:

$$\text{Adjusted earnings} = \text{earnings} + \text{net interest payments} - R \cdot K,$$

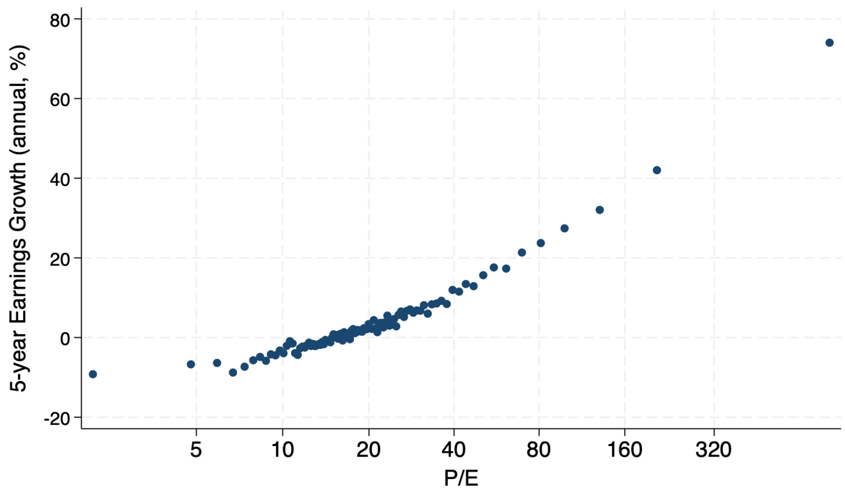
where K is the book value of assets and R is set to 0.10. The goal is to remove payments to physical capital so that the adjusted earnings better reflect rents to innovation.

We likewise adjust market capitalization as follows:

$$\text{Adjusted market cap.} = \text{market cap.} + \text{book value of net debt} - K.$$

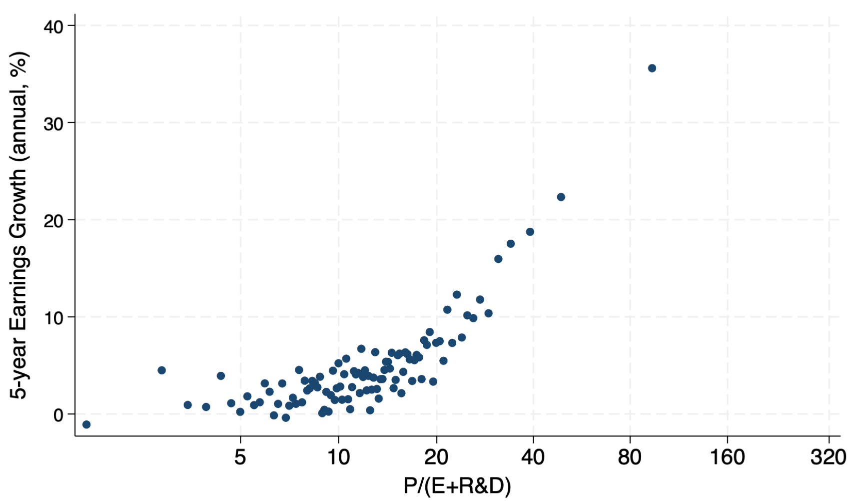
The rationale is to remove the value of physical capital to get closer to market value of future rents from innovation.

Figure A12: Adjusted earnings growth vs. P/E, all industry



Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the P/E ratio in year t. The y-axis is average yearly earnings growth from year t to $t + 5$. Both variables are residualized on a cubic of the firm-specific discount rate from David et al. (2022), fiscal-year end month and year fixed effects. Each of P and E is adjusted for debt and physical capital to get closer to rents and the market value of rents from innovation.

Figure A13: Adjusted earnings growth vs. $P/(E + R\&D)$, manufacturing



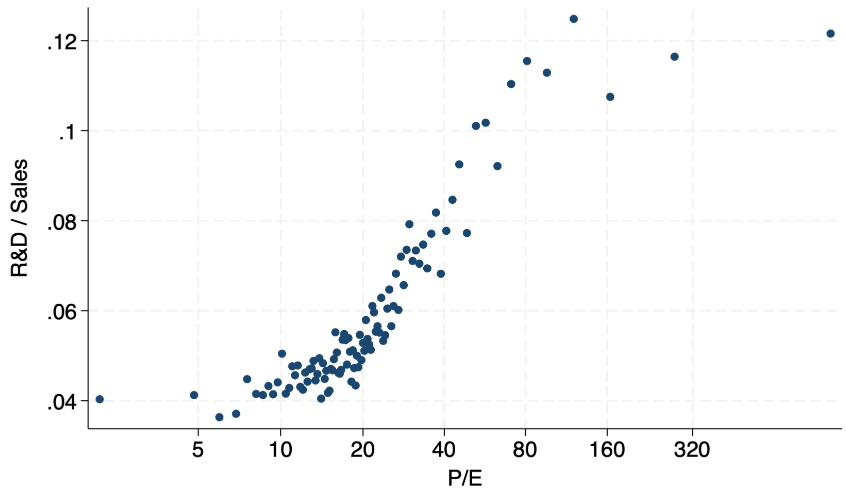
Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the $P/(E + R\&D)$ ratio in year t. The y-axis is average yearly earnings growth from year t to $t + 5$. Both variables are residualized on a cubic of the firm-specific discount rate from David et al. (2022), fiscal-year end month and year fixed effects. Each of P and E is adjusted for debt and physical capital to get closer to rents and the market value of rents from innovation. Manufacturing firms only.

Table A4: Adjusted growth of earnings and earnings/sales in manufacturing

	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$
g_E on $\log(P/(E + R\&D))$	0.265	0.142	0.099	0.077	0.067
	(0.012)	(0.007)	(0.006)	(0.005)	(0.005)
$g_{E+R\&D}$ on $\log(P/(E + R\&D))$	0.194	0.117	0.087	0.069	0.060
	(0.008)	(0.005)	(0.004)	(0.003)	(0.003)
$g_{(E+R\&D)/S}$	0.125	0.068	0.047	0.035	0.030
	(0.007)	(0.004)	(0.003)	(0.003)	(0.002)
g_S	0.069	0.049	0.040	0.034	0.030
	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)

Notes: Data from CRSP, I/B/E/S, and Compustat. Manufacturing firms only. The first row displays the coefficient from regressing average yearly earnings growth from year t to $t + h$ on the $\log P/(E + R\&D)$ ratio in t. The second row uses growth in earnings gross of R&D. The last two rows decompose growth in earnings gross of R&D into growth in the ratio of earnings gross of R&D to sales and growth in sales. Each column corresponds to a horizon of h years, $h = 1, \ldots, 5$. All regressions control for year and fiscal-year end month fixed effects, as well as a cubic in the firm-specific discount rate from David et al. (2022). Each of P and E is adjusted for debt and physical capital to get closer to rents and the market value of rents from innovation.

Figure A14: R&D/Sales vs. Adjusted P/E, manufacturing



Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the P/E ratio in year t. The y-axis is R&D over sales in year t. Both variables are residualized on a cubic of the firm-specific discount rate from David et al. (2022), fiscal-year end month and year fixed effects. Each of P and E is adjusted for debt and physical capital to get closer to rents and the market value of rents from innovation. Manufacturing firms only.

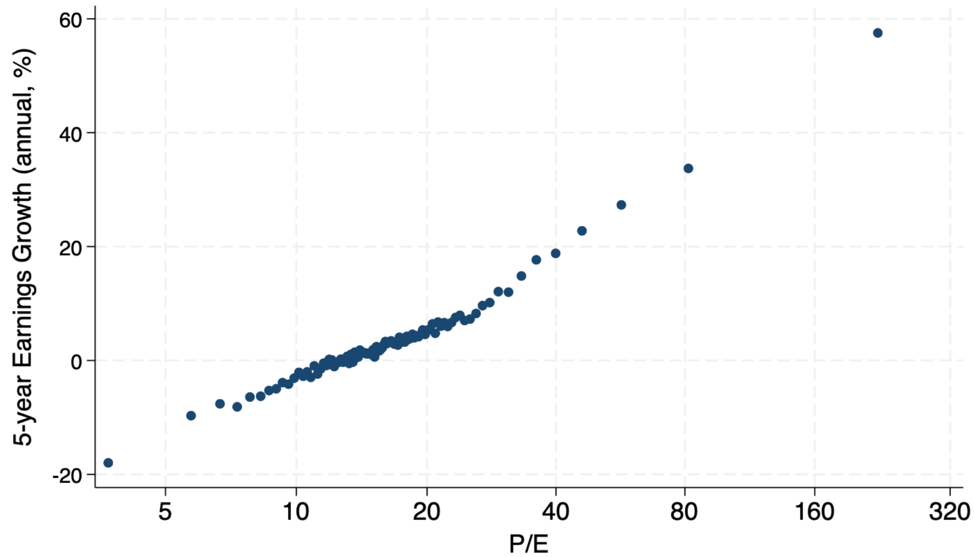
A.4 Controlling for firm fixed effects

In the main text, we control for firm-specific risk by residualizing on a measure of firm discount rate. Here we implement an alternative strategy by controlling for firm fixed effects. We control for firm fixed effects via

$$\log{(P/E)}_{i,t} = \alpha_i + \tau_{y(t)} + \mu_{m(t)} + \varepsilon_{i,t}$$

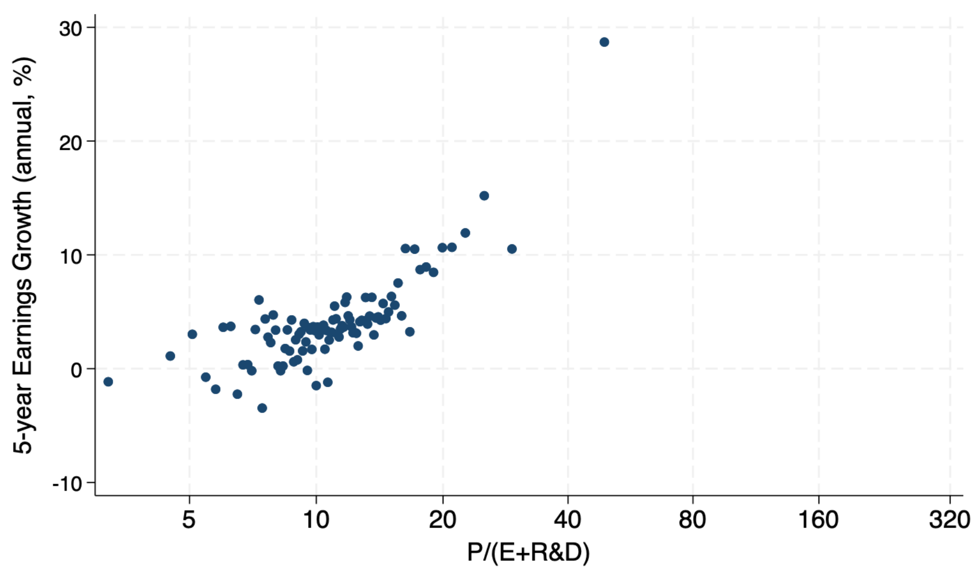
where α_i is a firm fixed effect, $\tau_{y(t)}$ a year fixed effect, and $\mu_{m(t)}$ a fiscal-year end month fixed effect, and collect the residuals $\varepsilon_{i,t}$.

Figure A15: Earnings growth vs. P/E, all industry



Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the P/E ratio in year t. The y-axis is average yearly earnings growth from year t to $t + 5$. Both variables are residualized on firm, fiscal-year end month, and year fixed effects.

Figure A16: Earnings growth vs. $P/(E + R\&D)$, manufacturing



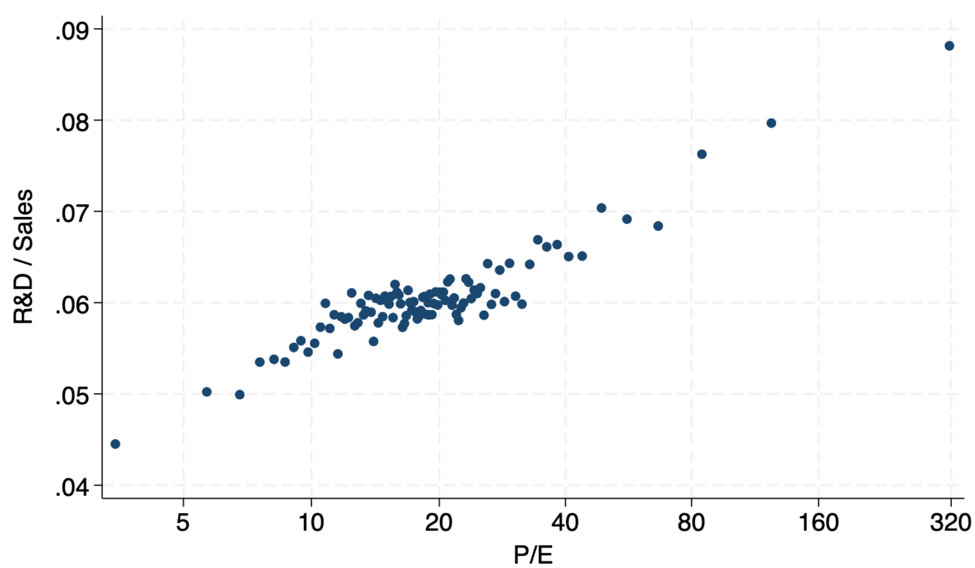
Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the $P/(E{+}R\&D)$ ratio in year t. The y-axis is average yearly earnings growth from year t to $t{+}5$. Both variables are residualized on firm, fiscal-year end month, and year fixed effects. Manufacturing firms only.

Table A5: Growth of earnings and earnings/sales in manufacturing

	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$
g_E on $\log(P/(E + R\&D))$	0.541	0.242	0.153	0.105	0.079
	(0.027)	(0.015)	(0.012)	(0.010)	(0.010)
$g_{E+R\&D}$ on $\log(P/(E + R\&D))$	0.358	0.198	0.134	0.098	0.079
	(0.010)	(0.006)	(0.005)	(0.004)	(0.004)
$g_{(E+R\&D)/S}$	0.248	0.128	0.083	0.059	0.048
	(0.009)	(0.006)	(0.004)	(0.004)	(0.003)
g_S	0.110	0.070	0.050	0.038	0.032
	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)

Notes: Data from CRSP, I/B/E/S, and Compustat. Manufacturing firms only. The first row displays the coefficient from regressing average yearly earnings growth from year t to $t + h$ on the $\log P/(E + R\&D)$ ratio in t. The second row uses growth in earnings gross of R&D. The last two rows decompose growth in earnings gross of R&D into growth in the ratio of earnings gross of R&D to sales and growth in sales. Each column corresponds to a horizon of h years, $h = 1, \ldots, 5$. All regressions include firm, fiscal-year end month, and year fixed effects.

Figure A17: R&D/Sales vs. P/E, manufacturing



Notes: Data from CRSP, I/B/E/S, and Compustat. The figure is a binned scatterplot with 100 bins. The x-axis is the P/E ratio in year t. The y-axis is R&D over sales in year t. Both variables are residualized on firm, fiscal-year end month, and year fixed effects. Manufacturing firms only.

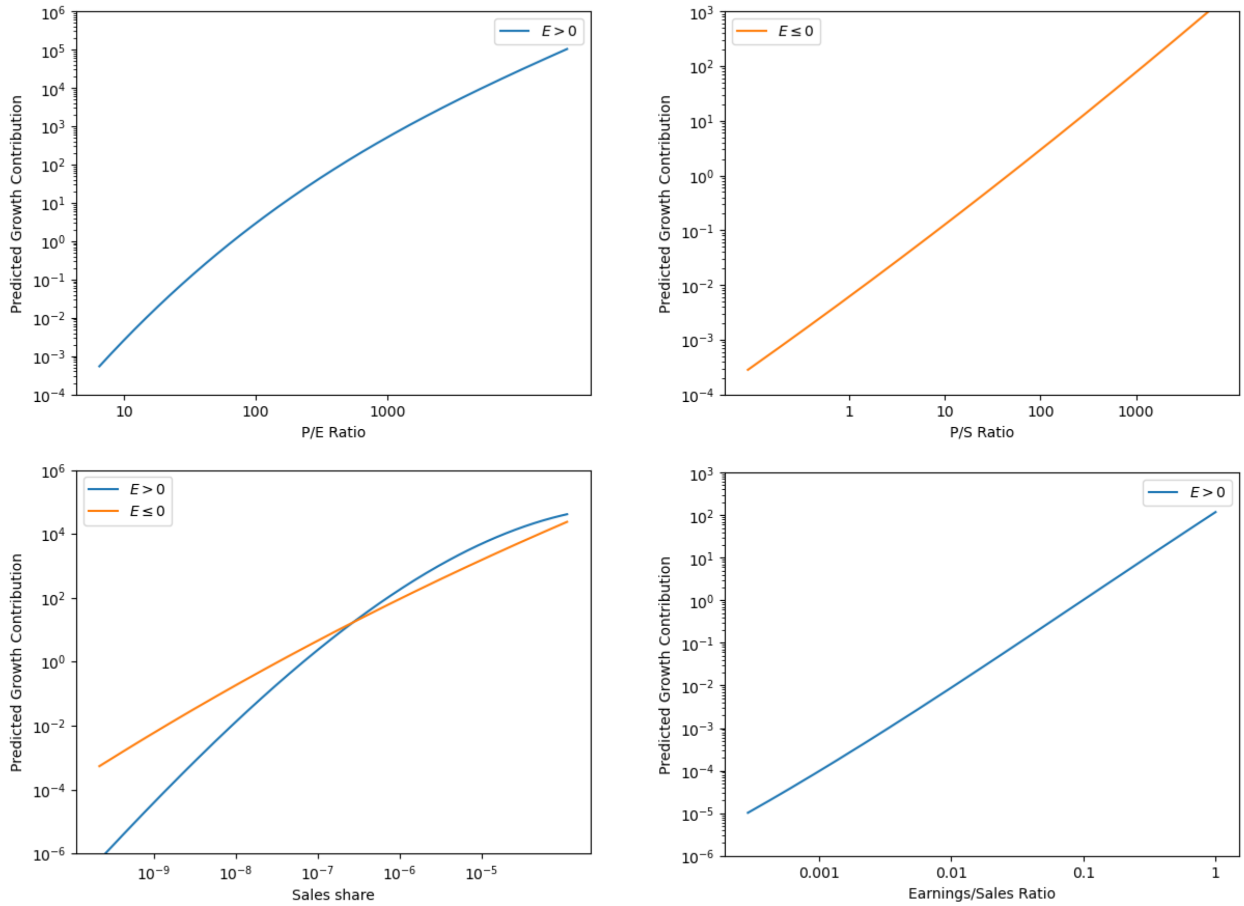
Table A6: Calibration target moments under alternative controls and adjustments

Target	Baseline	No risk control	Firm FE control	Age control	Capital & debt adjusted
1.	1.2%	1.2%	1.2%	1.9%	2.5%
2.	9.7%	9.7%	9.7%	9.6%	10.9%
3.	43%	43%	60%	52%	55%
4.	96%	91%	100%	100%	83%
5.	0.12	0.12	0.12	0.12	0.10
6.	0.09	0.09	0.07	0.08	0.09
7.	0.15	0.15	0.14	0.13	0.15
8.	-0.028	-0.028	-0.028	-0.019	-0.018
9.	17.5	17.5	17.5	14.7	17.0
10.	0.64	0.64	0.44	0.63	0.89

Notes: 1. Growth in aggregate earnings, net of labor input growth, $g^\star$. 2. Average firm-specific discount rate, $r^\star$. 3. Contribution of $g_{E/S}$ to g_E and log $P/(E + R\&D)$ relationship. 4. Contribution of $g_{(E+R\&D)/S}$ to $g_{E/S}$ and log $P/(E + R\&D)$ relationship. 5. Aggregate earnings/aggregate sales. 6. 90–10 range of fitted g_E. 7. 90–10 range of fitted $g_{E+R\&D}$. 8. R&D/sales projected on log sales. 9. 90:50 sales ratio. 10. Standard deviation of log $P/(E + R\&D)$.

B Model

Figure A18: Predictors of expected growth contribution



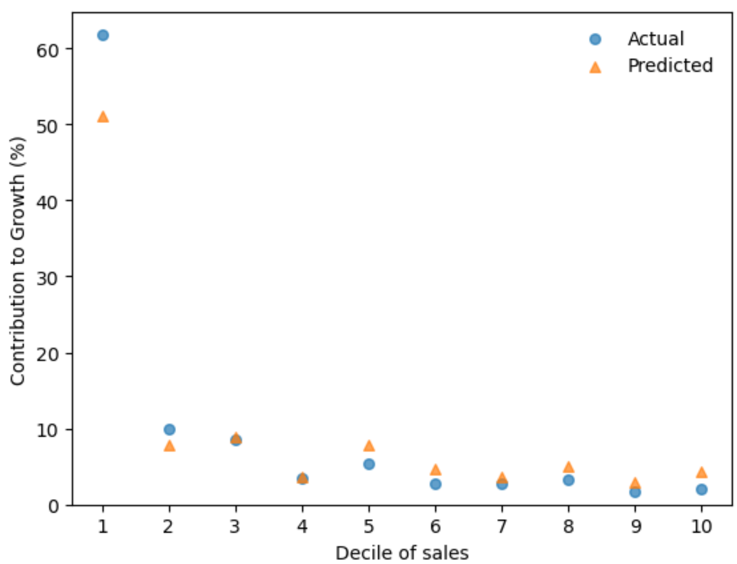
Notes: The figure shows how predicted growth contribution varies with the variable on the x-axis, relative to the predicted contribution at the mean of the variable.

Table A7: Marginal R^2 of expected growth contribution predictors in the model

Excluded term	ΔR^2
$\log(P/E) \times \mathbb{1}\{\text{Earnings} > 0\}$	0.50
$\log(P/S) \times (1 - \mathbb{1}\{\text{Earnings} > 0\})$	0.01
$\log(\text{Sales}) \times \mathbb{1}\{\text{Earnings} > 0\}$	0.49
$\log(\text{Sales}) \times (1 - \mathbb{1}\{\text{Earnings} > 0\})$	0.02
$\log(\text{Earnings/Sales}) \times \mathbb{1}\{\text{Earnings} > 0\}$	0.54

Notes: Full model R^2 is 0.71. The column ΔR^2 shows the decline in the R^2 when the listed term is excluded.

Figure A19: Actual contribution to growth by sales decile, the model



Notes: Model simulation. The figure shows the actual growth contribution of firms in each sales decile, along with the contribution predicted by the proxy relationship described in the main text.